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Review article

Graph neural networks to model and optimize the operation of Water Distribution Networks: A review

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ABSTRACT

Water Distribution Networks (WDNs) have become increasingly complex and interconnected, and the need for advanced modeling and optimization techniques has become fundamental to ensure an efficient and reliable clean water supply. Representing WDNs as graphs naturally models the underlying interacting physical structure and enables the usage of Graph Neural Networks (GNN) that combine the physical structure with abstract notions to capture local and global relationships. GNNs offer significant advantages in contrast to generic Deep Learning (DL) techniques and stand out as a promising solution to model intricate dependencies and enable the investigation of key challenges such as leak detection, water quality monitoring, and demand forecasting. This review presents the physics and hydraulics involved in WDN and the prevalent graph-based models used in the literature. The theoretical foundations of GNNs are shown, highlighting their capabilities in capturing complex spatial relationships and dependencies inherent in the network topology. The most promising GNN-based solutions that can address some of the most critical challenges of WDNs are discussed in detail. We outline the open challenges and potential directions for future developments in this field. By combining multidisciplinary and real-world aspects, this critical review highlights the role of GNNs in modeling and optimizing WDNs.

1. Introduction

A Water Distribution Network (WDN) is an organized network of pipes, pumps, valves, storage tanks, and other infrastructure designed to distribute potable water from a water treatment plant or other water sources to consumers. Such networks play a crucial role in ensuring a reliable and efficient supply of clean drinking water to houses, businesses, and various institutions within a community [1,2]. However, global water consumption is on a steady rise due to population growth and urbanization, experiencing a substantial annual increase in demand, juxtaposed with a looming global water shortage predicted to reach about 40% by 2030 [3,4]. Additionally, changes in temperatures, storms, and rainfall patterns require better management of water distribution across regions and seasons.

Despite efforts to optimize Water Supply Systems (WSSs) for performance, economy, and equity to provide Continuous Water Supply (CWS), decreasing supply and increasing demands, in combination with high leakage levels and asset failure, primarily attributed to outdated infrastructure, often lead to either a complete collapse of service or partial service in parts of the network for limited periods, resulting in Intermittent Water Supply (IWS). IWS frequently catches operators unprepared, leading to abnormal and heavy demand. This issue is further exacerbated by failures in critical system components such as pipes and pumps, high leakage rates (Non-Revenue Water, NRW), and deterioration in water quality due to pollutant infiltration during unpressurized periods. Aging infrastructure and insufficient investment in maintenance and rehabilitation contribute to these challenges, with recent statistics reporting that water loss can reach an average of 26% in Europe, or even exceed 50% in some regions and seasons [5].

Efficient modeling of a WDN is essential for optimizing operation and ensuring a sustainable and reliable water supply while minimizing water loss and optimizing resource use [6]. Advances in Information and Communication Technologies (ICTs), particularly the Internet of Things (IoT) and 5G, combined with artificial intelligence (AI) and deep learning (DL), are driving the digitization of WDNs. Information and Communication Technologies (ICTs) and, in particular, the Internet of Things (IoT) and 5G technologies, in combination with advanced data analysis using artificial intelligence (AI), can lead to the digitization of WDNs. The densification of sensor networks, the ever-increasing

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number of connected devices, and the transition towards new efficient wireless technologies offer a unique opportunity to monitor the WDNs' operation continuously. The availability of such data, combined with recent deep learning (DL) developments, can significantly increase the possibility of addressing specific key challenges for WDN. IoT data collected and analyzed can dramatically enhance system modeling and help optimize the operation, for example, (i) to predict pressure peaks that compromise efficiency and reliability and cause significant strain on the infrastructure, leading to potential damage and service disruption [7]; (ii) optimize supply based on the ever-evolving user needs, maintain a steady quality of water flow and avoid areas of over or under-supply [8]; and/or (iii) efficiently and on-time detect leaks to maintain the integrity and functionality of the WDN [9].

The ability of DL models to learn intricate patterns and dependencies between data has been proved efficient in refining the accuracy and reliability of these tasks' resolution. However, moving from laboratory to real-world deployments poses particular difficulties. On the one hand, deploying sensors to monitor the operation of the entire WDN incurs very high costs, increases the operation complexity, and amplifies the operators' need to use the limited radio spectrum resources efficiently. On the other hand, the trend of deploying cloud-centric domain-agnostic AI models with increasing complexity of the training techniques in combination with high data volumes used further increases the cost of the ICT infrastructures. For these reasons, despite the benefits mentioned above, today, the vast potential of ICT-enabled WDN remains untapped, mainly by the consumer market, and the work conducted by academia and industry has primarily focused on isolated aspects of these systems.

In the last decade, Graph Neural Networks (GNN) have been introduced as a particular type of DL, which is when the underlying data can be represented as a graph. In contrast to generic DL methods, these methods are superior in learning graph-structured data. WDN are naturally represented as graphs, with nodes corresponding to components like junctions or reservoirs and edges representing connections like pipes. Representing WDNs as graphs offers significant advantages, enabling the modeling of physical or abstract components as a graph and capturing local and global relationships within the network. Such an abstract structure makes GNNs an ideal choice for modeling these systems. Unlike DL models, GNNs capture local and global relationships by considering hydraulic models that model the complex dependencies within WDNs, such as flow and pressure dynamics across the network [10,11]. By leveraging network topology and associated physical data, GNNs have strong potential in tasks like leak detection, nodal pressure reconstruction, and demand forecasting. Their ability to generalize across diverse network configurations and learn from limited data makes them especially valuable for real-world applications. Graph-based methods, particularly suitable in the WDN context, have been developed to characterize networked behavior naturally associated with these systems [12]. More recently, a new branch of signal processing known as Graph Signal Processing (GSP) has been developed [13] to study signals defined on graphs, i.e., signals that lie on interconnected domains. The GSP modeling approach matches the WDN context, where the network can be represented as a graph, and physical quantities, such as pressure and water flow, can be modeled as graph signals. Such modeling can be used to develop innovative methods for monitoring WDN [14] or detecting leaks [15].

Positioning within industrial information integration. Recent work in Industrial Information Integration has highlighted the benefits of GNNs of intelligent modeling and operation of cyber–physical infrastructure systems. In particular, applying GNNs to WDNs exemplifies a broader trend in industrial digital transformation — integrating structural domain knowledge, sensor networks, and data-driven optimization under a unified modeling framework.

Several recent studies published emphasize the growing role of graph-based deep learning for industrial applications. For example, GNNs have been applied to enhance link prediction through subgraph structures [16], and to improve real-time anomaly detection in healthcare-oriented industrial settings using masked graph neural networks [17]. Furthermore, contrastive graph transformers have shown promise in spatio-temporal prediction tasks, such as energy demand forecasting for sustainable manufacturing [18].

Beyond GNNs, symbolic graph-based approaches such as knowledge graphs have been actively explored to support intelligent reasoning and industrial data integration. For instance, knowledge graph embeddings have been used in meta-learning frameworks for defect inspection and decision support [19], and ontology-enhanced knowledge graphs have enabled model management for systems engineering workflows [20]. Notably, recent work on industrial dataspace–based knowledge graph construction for smart maintenance [21] demonstrates how semantic graph structures can serve as a foundation for intelligent services in cyber–physical environments.

By synthesizing recent developments in GNN architectures and applying them to domain-specific challenges in WDNs, such as leak detection, demand forecasting, and sensor placement, this work aims to situate itself within the emerging integrated, intelligent industrial infrastructure field. It bridges hydraulic modeling and broader industrial information integration efforts that leverage graph-structured learning for smarter infrastructure management.

Differences between this survey and existing ones. Several recent surveys have explored theory and methods for WDN management from different perspectives, focusing typically on broader theoretical aspects or more specific application scenarios. For instance, Kashid et al. [22] review the evolution of intelligent systems for water management, emphasizing real-time data collection and forecasting. Similarly, Chan et al. [23] discuss methodologies such as support vector machines and neural networks (NN) for leakage detection without considering the data's graph structure. A presentation of data-driven results for leak detection is given in the work of Kammoun et al. [24], where authors present comparisons of machine learning (ML) techniques on benchmark datasets but lack insights into the structural representation of WDNs as graphs. At the same time, El et al. [25] categorize leak detection approaches into static and dynamic without integrating recent advancements using GNNs. *In contrast to these surveys, we focus on integrating GNNs with WDN challenges and examine graph-based methods or advanced learning approaches.* More recently, some surveys look into the combination of IoT and ICT technologies to collect, transmit, and process the status of the WDN, such as one by Pagano et al. [26] that mainly focuses on scalability and data collection challenges in large-scale deployments. Rathi et al. [27] provides an in-depth review of sensor placement methods but remains limited to contamination detection objectives, leaving other critical aspects of WDN management unexplored. In contrast to these surveys, *we do not focus on the technological infrastructure to sense the current status of the components of the WDN and transmit the data for processing.*

The survey presented in this paper distinguishes itself from existing literature reviews on GNNs [28,29] by its domain-specific focus. While other surveys provide comprehensive overviews of graph-based learning techniques, they primarily emphasize applications such as traffic forecasting [30], routing optimization [31], social recommender systems [32], intrusion detection [33] or molecular structures [34,35], with limited attention to WDNs. In contrast, *our work examines studies that specifically address the utilization of GNNs in solving problems related to WDNs*. This targeted approach ensures a comprehensive analysis of methodologies, challenges, and advancements pertinent to this critical infrastructure domain. Our survey differs from general surveys on DL applications on WDN problems [36–38] *by concentrating exclusively on the unique contributions and capabilities of GNNs*. This distinction is crucial as GNNs offer specialized advantages in handling graph-structured data inherent in WDN, such as network topology modeling. We provide an in-depth exploration of GNN-specific strategies and their

efficacy in enhancing the performance and reliability of WDNs, thereby filling a significant gap in the existing body of research. Moreover, we give an overview of theoretical basics and the most known challenges within WDNs, showing how researchers approached them with GNN methodologies and highlighting these techniques' advantages and future directions.

This survey aims to explore the theory and applicability of some advanced Graph Neural Network (GNN) methodologies in the design, monitoring, and optimization of the operation of WDN. In Section 2, we present a detailed background on the physical properties of WDNs, that is, the hydraulic models and governing laws related to pipe flows. This critical information is often neglected in computer science, while they are fundamental for thoroughly investigating the problem. Then, Section 3 proceeds by looking into how the physical laws governing WDNs are modeled in the recent and relevant bibliography using graph theory. Section 4 introduce GNNs, explaining their theoretical background and how the WDN properties fit on specific GNN architectures already applied in the context of WDN. Next, in Section 5, we review the available datasets and the key characteristics and provide an overview of WDN modeling tools that can produce synthetic datasets. Given all the above information, in Section 6, we present how GNNs have been used to address crucial challenges in the WDN domain, such as optimal sensor placement, leak detection, and demand forecasting. Finally, in Section 7, we discuss future research and open issues in this domain, and the paper concludes with Section 8. This work aims to present and discuss the potential of graph-based learning in WDNs for more efficient and sustainable management of water resources.

2. Physical properties of WDNs

A short introduction to the operating principles of WDNs is presented here. Those interested in an in-depth introduction are referred to [39,40]. The key components of a WDN are depicted in Fig. 1 and include

1. *junctions* where links join and where water enters or leaves, e.g., to deliver water to individual consumers in a utility's service area in the required quantity, quality, and pressure;
2. *reservoirs*, external source of water such as a lake;
3. *storage tanks* required to meet fluctuation in demand in emergencies, such as fire fighting, or to balance storage to reduce energy cost;
4. *pumps* that provide the necessary energy to overcome variations in elevation and compensate for energy dissipation due to friction;
5. *pipes* that connect the different components of the network and convey water from one point to another;
6. *emitters* are devices associated with junctions that measure outflow through an orifice, e.g., used to detect leakages;
7. *valves* that are flow-control devices, such as regulating or pressure-reducing valves, limit the flow pressure at specific points in the network.

The design goal of a WDN is to provide continuous water supply with appropriate pressure under various *loading conditions*, which are spatial patterns of *demands* defining the flow requirements for a user. The *flow rate*, the volume or mass of fluid that passes per unit of time, is one of the variables that describe hydraulic conditions. A fluid flowing in conduits experiences a complex exchange of mechanical energy; the work associated with the movement relates to changes in gravitation potential energy and kinetic energy. Energy can also be lost due to friction, causing negligible increases in temperature. For historical reasons, we refer to *heads* instead of energies, associating them to vertical distances, listed in Table 1.

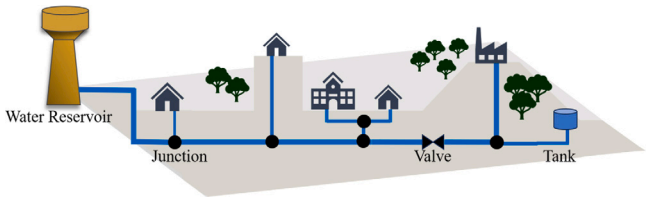


Fig. 1. Graphical representation of the main components of a WDN.

Table 1
Types of heads in fluid mechanics, where p refers to pressure, γ to the specific weight of the fluid, Z to the elevation w.r.t. a certain point, v to the velocity and g to gravity acceleration.

Head	Definition	Description
Pressure head	$\frac{p}{\gamma}$	Flow work
Elevation head	Z	Gravitational potential energy
Velocity head	$\frac{v^2}{2g}$	Kinetic energy
Piezometric head	$\frac{p}{\gamma} + Z$	Pressure + elevation head

2.1. System state and governing laws of pipe flow

The system state can either be *steady* or *unsteady*, contingent upon the constancy or variation of the loading conditions. When subjected to constant loading, the system is in a steady state, characterized by the absence of varying time flow in pipes and pressure heads at nodes. On the contrary, unsteady conditions prevail when the system's state fluctuates over time. The heads are usually computed at the junctions connecting the water withdrawal point from the network. The nodes of the WDN where the total energy is known, such as a tank, are called fixed-grade nodes.

The governing laws for pipe flow under steady-state conditions in a WDN are:

- *Conservation of Mass*: the storage rate equals the difference between inflow and outflow to the system. For a junction node

$$\sum Q_{in} - \sum Q_{out} = Q_{ext} \tag{1}$$

  where Q_{in} and Q_{out} are the inflows and outflows through connected pipes, and Q_{ext} is the demand.
- *Conservation of Energy*: for any path between two points, the energy difference equals the energy added to the flow by pumps and the frictional and minor losses. Expressing energy ϵ in terms of heads:

$$\sum H_{\text{loss},i} + \sum H_{\text{pump},j} = \Delta\epsilon \tag{2}$$

  where $H_{\text{loss},i}$ is the head loss associated with the component i, and $H_{\text{pump},j}$ is the added head of pump j.

The hydraulics of the WDN are calculated based on the above expressions for all nodes and pipes. In this way, a system of nonlinear equations can be linearized under further assumptions, such as knowing the boundary conditions and the nodal demands. The solution can be found by iterative numerical methods, for example, using a hydraulic simulator such as EPANET [41], see also Section 5.1.

Under unsteady conditions with the rigid pipe assumption, the law of conservation of momentum states that the sum of forces acting on the volume of fluid in a pipe is equal to the time rate of change of momentum:

$$\sum F = F_1 - F_2 - F_f = \frac{d(mv)}{dt} \tag{3}$$

where F_1 and F_2 are forces on the ends of the pipe element, F_f is the frictional force, and mv is the momentum of the fluid.

2.2. Head loss and frictional losses

The head loss associated with frictional losses in pipes is the difference in the total head from the beginning to the end of a pipe over a distance l. The Darcy–Weisbach and Hazen–Williams equations are the most commonly used head loss formulas. The Darcy–Weisbach equation is defined as:

$$H_{D-W} = f\frac{l}{d}\frac{v^2}{2g} \tag{4}$$

where f is the friction factor, l is the pipe length, d is the pipe diameter and v is the mean flow velocity. The Hazen–Williams equation is defined as:

$$H_{H-W} = \frac{10.74l}{d^{4.86}}\left(\frac{Q}{C_{H-W}}\right)^{1.85} \tag{5}$$

where C_{H-W} is the Hazen–Williams roughness coefficient, Q is the flow, l is the pipe length, and d is the pipe diameter. The same equation can be expressed using feet as a unit of measurement as

$$H_f = \frac{4.72l}{D^{4.86}}\left(\frac{Q}{C_{H-W}}\right)^{1.85} \tag{6}$$

Head loss may also occur due to changes in velocity or direction, called local head loss, which is expressed as:

$$H_{\text{minor}} = k\frac{v^2}{2g} \tag{7}$$

where k is the minor loss coefficient.

2.3. Hydraulic modeling of WDNs

One key hydraulic property of WDNs is the relationship between pressure and demand at nodes. This relationship is often modeled using the Pressure-Driven Analysis (PDA), which calculates the actual water output (demand) based on the pressure at each node. The formula is given by:

$$D_i^* = \begin{cases} 0, & p_i < p_i^{min} \\ D_i\left(\frac{p_i - p_i^{min}}{p_i^{req} - p_i^{min}}\right)^{\delta}, & p_i^{min} < p_i < p_i^{req} \\ D_i, & p_i > p_i^{req} \end{cases} \tag{8}$$

where D_i^* is the actual demand, p_i is the pressure at node i, p_i^{min} is the minimum pressure for water delivery, and p_i^{req} is the required pressure for full delivery. The exponent δ is typically set to 0.5. This formula allows for modeling how nodal demand fluctuates with pressure, providing a basis for pressure-demand analysis. In Demand-Driven Analysis (DDA), the network provides the required demand at each node, regardless of the available pressure. The assumption is valid when the network is well-pressurized to meet all demands. However, this sometimes results in hydraulic solutions with negative pressures that are physically impossible.

2.4. Pipe burst flow rate

In the event of pipe bursts, WDNs must account for the burst flow rate, which can be modeled using hydraulic principles. The burst flow rate is determined by the Bernoulli equation and expressed as:

$$Q_{burst} = \beta\mu A_d\sqrt{2gH_p} \tag{9}$$

where Q_{burst} is the burst flow rate, β is a coefficient (usually 0.67), μ is the flow coefficient (typically 0.61), A_d is the cross-sectional area of the pipe connected to the burst node and H_p is the pressure head at the node. In practical applications, the emitter coefficient c is used to simplify the burst flow rate calculation in hydraulic simulations. The simplified formula is:

$$Q_{burst} = cH^{\delta} \tag{10}$$

where c is the emitter coefficient and δ is the pressure index, typically set to 0.5. Additionally, the leakage demand is expressed as:

$$D_{\text{leak}} = \mu A_{\text{leak}} p^{\alpha}\sqrt{\frac{2}{p}} \tag{11}$$

where μ is the flow coefficient, A_{leak} is the leakage area, p is the pressure, and α is used to represent leakage characteristics.

3. Graph theory-based modeling of WDNs

We provide a brief but self-contained summary of the most basic definitions in graph theory as a single place for easy reference of the fundamental terms based on the terminology of [42]. Then, we introduce different approaches to how graph theory can be used to model the WDNs.

3.1. Basic terminology

A graph $\mathcal{G} = (V, E)$ is a pair of finite sets satisfying $E \subseteq [V]^2$; thus, the elements of E are two-element subsets of V. The elements of V are the *vertices* (or *nodes*) of the graph $\mathcal{G}$, and the elements of E are its *edges* (or *links*). A vertex u is *incident* with an edge e if $u \in e$, and e is said to *join* its two *end vertices*. An edge $\{x, y\}$ is denoted as xy. Two vertices u, v are *adjacent* (or *neighbors*) if uv is an edge of $\mathcal{G}$. The *neighborhood* of a vertex v, denoted as $\mathcal{N}(v)$, consists of all nodes adjacent to v, and the *degree* of v is given by $|\mathcal{N}(v)|$. If two edges $e \neq f$ share a common vertex, they are called *adjacent edges*.

Considering that $|V| = N$ and $|E| = N_e$, the adjacency matrix $\mathbf{A}$ of the graph $\mathcal{G}$ is a $N \times N$ matrix that encapsulates interactions between nodes. Each element a_{ij} of the matrix is set to zero if there is no link connecting node i to node j. The adjacency matrix can either be binary or weighted. The weight assigned to a link depends on the specific application; it could be the correlation coefficient between signals at two nodes or physical properties associated with the specific component of the WDN, e.g., the diameter of the pipes. The *degree matrix* $\mathbf{D}$ is a diagonal matrix where each entry $d_{ii} = \sum_{j=1}^{N} a_{ij}$ represents the degree of node i. The *graph Laplacian*, defined as

$$\mathbf{L} = \mathbf{D} - \mathbf{A},$$

is a key matrix in spectral graph theory, capturing structural properties of the network.

Since we focus on GNNs for WDNs, it is essential to define graph-based data representations relevant to WDNs. Each node in the graph represents a physical component of the WDN (e.g., a junction, a reservoir, or a pump), and edges represent physical connections (e.g., pipes). A *graph signal* is a function that assigns a value to each node or edge, representing system properties such as pressure, flow rate, or water quality at a given time. Node features, denoted as $\mathbf{X} \in \mathbb{R}^{N\times d}$, with $x_v \in \mathbb{R}^d$ representing the feature vector of a node v, encapsulate information like demand, pressure, or sensor readings. Edge features, denoted as $\mathbf{X}^e \in \mathbb{R}^{N_e\times c}$, with $x^e_{v,u} \in \mathbb{R}^c$, can describe pipe characteristics such as length, diameter, or roughness. Additionally, a graph may have graph-level attributes that encapsulate properties or features describing the entire graph, such as global structural properties, graph labels, or other global characteristics. These graph representations serve as inputs to GNN models, where additional matrix normalizations and transformations (such as normalized adjacency or Laplacian matrices) may be introduced later to ensure stable learning and practical information propagation.

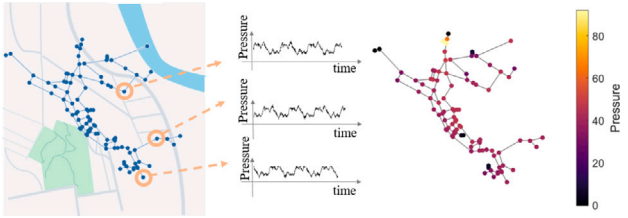


Fig. 2. Representation of how to derive the graph abstracting junctions and pipes. On the left, the WDN of Net3 (see Table 2) with pipes and junctions is represented on the territory map. On the right, we have the associated graph where nodes correspond to junctions and links to pipes. In this case, the attribute associated with each node is the pressure (m) represented across time for the junctions circled in orange. Each node is colored according to its pressure value.

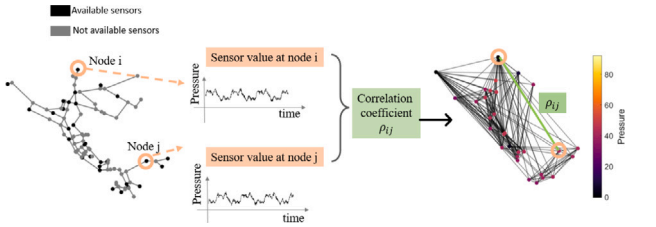


Fig. 3. Representation of how to derive the graph abstracting sensing devices. On the left, we have the territory map of WDN of Net3 (see Table 2), where pipes and junctions monitored by functional sensors are colored in black, while all other nodes where no sensor is available are colored in gray. On the right, we have the associated graph where nodes correspond to sensors and links represent aggregated connections based on the available sensor data.

3.2. Graph abstracting junctions and pipes

The first approach of using graph theory to model a WDN is following a simple assignment where nodes represent junctions. The edges represent the pipes that convey water from one point to another. An edge connects two nodes of the graph if a pipe connects the junctions of the WDN.

More formally, if the junctions represented by nodes i, j are connected via a pipe, then the edge $e_{ij} \in E$. Attributes can be assigned to the graph nodes to introduce additional information regarding, for example, the elevation of the junction from sea levels (in meters) or the required demand (in L/s). Weights may be assigned to edges to represent the length of the pipe (m), the diameter (mm), or the hydraulic loss, like in the case of [43]. For an example of how the graph is constructed, see Fig. 2. This basic modeling approach is the most used in the relevant literature for studying the problem of leak detection and localization presented in Section 6.2.

3.3. Graph abstracting sensing devices

In practical applications, not every junction within a WDN may be equipped with sensors due to economic constraints or physical inaccessibility in certain areas. Consequently, sensor deployment is limited to specific points in the network [44]. In such cases, only a subset of junctions, those with available sensor data, are modeled, and the edges connecting these junctions might not represent the actual physical pipes but abstract connections between the sensed junctions. This abstraction can result in a reduced or incomplete graph representation of the WDN where the graph edges do not necessarily correspond to actual physical pipes. Still, it may represent aggregated connections based on the available sensor data.

The sensing devices can measure different properties of the water flowing through the WDN, such as the hydraulic pressure [45] or the water quality. The measurement can be continuous or periodic, and the observations collected over time constitute a *time series* or *signal*. The time series of each sensing point are analyzed to identify correlations and characterize their similarity. If two time series have a high similarity, then the graph nodes corresponding to the sensing devices are connected by an edge.

Various techniques can be used to calculate the difference between two time series, for example, by examining the point-by-point difference using the Euclidean distance [46] or the mean absolute percentage error (MAPE) or by measuring the linear correlation using Pearson's correlation coefficient method [47]. More elaborate methods can be used to quantify the similarity across different time series that are more robust when the data series are slightly time-shifted, such as dynamic time warping [48] or compression-based dissimilarity [49].

Given the above approach, a WDN is modeled by a graph $\mathcal{G}$ where the sensing devices represent each node. We say that an $e_{ij} \in E$ if the time series of the observations of the sensing devices abstracted by nodes i, j correlate a certain threshold θ. An example of this process is depicted in Fig. 3 where the nodes of $\mathcal{G}$ correspond to the black nodes. Then, we compute the Pearson correlation coefficient between each node pressure value and the link weights, which are calculated with a threshold of 0.9.

3.4. Alternative graph modeling approaches

Alternative graph representations exist that can offer new perspectives and may help expand the use of GNNs in designing and managing WDNs. While they have not yet been widely explored, they have significant potential for future applications in GNN development for WDN analysis.

One possibility is to leverage the operational structure of the WDNs. Water utilities commonly divide WDNs into smaller isolated zones known as District Metered Areas (DMAs), where flow and pressure sensors are strategically placed — typically at entry points and supplemented with a few internal pressure sensors. This segmentation aids in leak detection and pressure regulation while minimizing sensor deployment costs [44]. A graph representation based on this operational structure could capture the inherent segmentation of the network, aligning with real-world management practices.

Another alternative representation used in the work by Cattai et al. [14] builds upon the line graph $\mathcal{G}_L = (V_L, E_L)$ which is related to the original graph $\mathcal{G}$. Here each node $v_l \in V_L$ of $\mathcal{G}_L$ corresponds to one link in $\mathcal{G}$ and two nodes in $\mathcal{G}$ are connected if the corresponding links $e_1, e_2 \in E$ of the original graph originate from a common vertex $v \in V$ [50–52]. The adjacency matrix $\mathbf{A}_L \in \mathbb{R}^{N_e \times N_e}$, can be derived from the adjacency matrix $\mathbf{A}$ as $\mathbf{A}_L = \mathbf{B}\mathbf{B}^T - 2\mathbf{I}$, where $\mathbf{B} \in \mathbb{R}^{N \times N_e}$ is the original graph incidence matrix, with its generic binary element b_{ij} is one if the node i is involved in the jth link. This transformation allows signals originally defined on edges of $\mathcal{G}$, such as hydraulic flows, to be modeled as node signals in the line graph. The latter is advantageous because it enables the application of graph signal processing techniques, which operate on node signals [13]. Graph signal processing techniques can be applied to reconstruct flows from a limited set of measurements by associating flow values with nodes in the line graph. For instance, Cattai et al. [14] employ this concept to develop a reconstruction algorithm for WDN monitoring when only a subset of edge measurements is available.

4. Types of graph neural networks

GNN is a neural network class designed to process graph data that leverages connectivity and feature information to capture the dependencies and relationships between entities. In this context, we call embeddings compressed feature representations that can be learned for vertices, edges, neighborhoods, or graphs. Most GNNs are based on an information diffusion process that involves smoothing features based on the graph's structure. For example, considering a scalar feature for

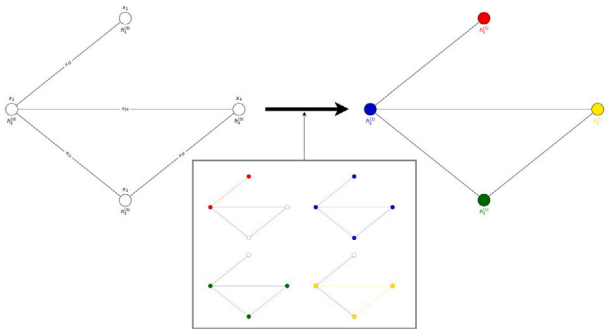


Fig. 4. RGNN single iteration example. The top left shows the initial graph features, the top right shows the updated embeddings and the bottom figure shows the quantities involved in the computation of each embedding.

each node $\mathbf{X} \in \mathbb{R}^N$, diffusion can be defined in terms of adjacency matrix as $\mathbf{X}' = \mathbf{A}\mathbf{X}$, corresponding to the node-wise operation $X_i' = \sum_{j \in \mathcal{N}(i)} A_{ij} X_j$. Diffusion can also be defined in terms of Laplacian, where one step can be defined as $[\mathbf{L}\mathbf{X}]_i = \sum_{(i,j) \in E} \mathbb{A}_{ij}(X_i - X_j)$ [53]. The simplest way to define a graph layer is to include the connectivity information to update the embeddings. Considering the node features, a graph layer can be defined as $\mathbf{H} = \mathcal{F}(\mathbf{X}, \mathbf{A})$, where $\mathcal{F}$ is a parametric function. GNN architectures can be broadly classified into Recurrent Graph Neural Networks (RGNN), Convolutional Graph Neural Networks (CGNN), Graph Autoencoders, and Spatio-Temporal GNN [29].

GNNs can address various analytics tasks that can be categorized based on the level of the graph elements involved: nodes, edges, or the entire graph. At the node level, GNNs can perform tasks like node classification and regression by propagating information across the graph and refining node representations through layers of convolution or aggregation. GNNs use the learned representations to predict edge attributes such as classification labels or connection strengths for tasks involving edges. At the graph level, GNNs generate a global graph representation for tasks like classification by aggregating node-level information using pooling and readout operations, followed by additional layers for final predictions [28,29].

4.1. Recurrent graph neural network

Recurrent Graph Neural Network (RGNN) is a graph-based model class that iteratively updates node representations through a shared parametric function by recurrently leveraging information from neighboring nodes. This iterative process continues until convergence or a predefined number of steps, enabling the model to capture long-range dependencies and complex interactions in graph-structured data. For a graphical representation see Fig. 4. RGNNs are particularly effective in scenarios where deep and dynamic information propagation is required, although their sequential nature can pose challenges for scalability and convergence. A simple approach to update the node v hidden state h_v at step k can be defined as follows:

$$h_v^{(k)} = \sum_{u \in \mathcal{N}(v)} \mathcal{F}(x_v, x_{vu}^e, x_u, h_u^{(k-1)}) \tag{12}$$

where $\mathcal{F}$ is a parametric function, h_v^0 are defined on initialization. The sum operation in aggregation removes neighbor count and order dependency. The expression defines a message-passing rule that allows information to propagate throughout the graph, one hop away at each iteration.

4.1.1. Gated graph neural network

A variant of RGNN is the Gated Graph Neural Network (GGNN), which implements a Gated Recurrent Unit (GRU), that is useful in performing tasks with long-term dependencies, i.e., when the repeating

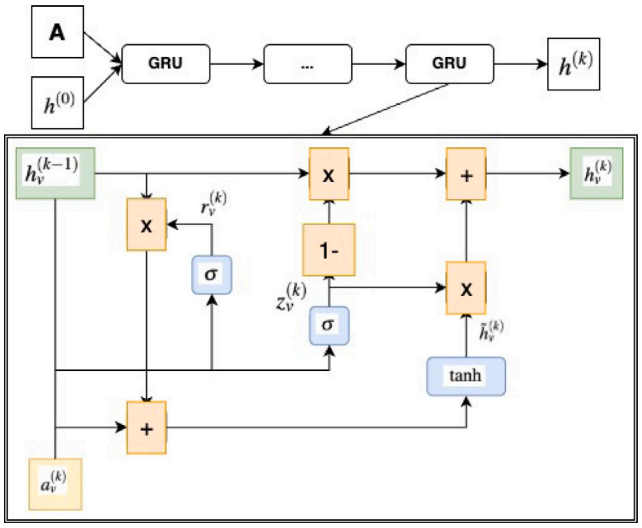


Fig. 5. Simplified diagram of a GGNN.

patterns are less frequent. To consider reverse connection in directed graphs, an extension of the adjacency matrix should be considered $\bar{\mathbf{A}} = [\mathbf{A}, \mathbf{A}^T] \in \mathbb{R}^{N \times 2N}$ as in [54], that might be not necessary in the case of undirected graphs. Node v at step k aggregates neighborhood information as follow:

$$a_v^{(k)} = \bar{\mathbf{A}}_{v:}[h_1^{(k-1)^T}, \ldots, h_N^{(k-1)^T}]^T + b \tag{13}$$

where $\bar{\mathbf{A}}_{v:} \in \mathbb{R}^{N \times 2}$ are the two columns of the block corresponding to node v and b is the bias vector. Then, GRU propagation steps to update node states can be described as follows:

$$r_v^{(k)} = \sigma(\mathbf{W_r} a_v^{(k)} + \mathbf{U_r} h_v^{(k-1)}) \tag{14}$$

$$z_v^{(k)} = \sigma(\mathbf{W_z} a_v^{(k)} + \mathbf{U_z} h_v^{(k-1)}) \tag{15}$$

$$\tilde{h}_v^{(k)} = \tanh(\mathbf{W} a_v^{(k)} + \mathbf{U}(r_v^{(k)} \odot h_v^{(k-1)})) \tag{16}$$

$$h_v^{(k)} = (1 - z_v^{(k)}) \odot h_v^{(k-1)} + z_v^{(k)} \odot \tilde{h}_v^{(k)} \tag{17}$$

where r and z are the reset and update gates, $\mathbf{W_r}$, $\mathbf{W_z}$, $\mathbf{W}$, $\mathbf{U_r}$, $\mathbf{U_z}$, $\mathbf{U}$ are the weights and biases, $\sigma(\cdot)$ is the sigmoid activation function and $\odot$ is the element-wise multiplication. GGNN computes recursively the state for K-step; each step considers information from neighbors 1 hop more away until it outputs a state matrix $h^{(K)} \in \mathbb{R}^{M \times S}$, where M is the graph size and S is the size of hidden states. The final output is then flattened into a vector and normalized into a probability distribution using a linear and a softmax layer (see Fig. 5).

4.2. Convolutional graph neural network

An extension of the convolution operator can be defined in the graph domain, which offers appealing properties such as locality and scalability [28]. Convolutions on graphs can be defined in two different domains: spatial and spectral. This categorization is no longer commonly used within the scientific community. However, we have decided to adopt it as it is still employed in most papers analyzed for this review.

4.2.1. Spatial approach

In a Spatial GCN, each node's embedding is updated by aggregating the embeddings of its neighbors. The latter is achieved by summing neighboring embeddings, applying a nonlinear activation function $\sigma(\cdot)$, and normalizing with a neighborhood scaling factor to prevent exploding values during the forward pass. Mathematically, for each node v at

layer k, the update rule is:

$$h_v^{(k)} = \sigma\left(\sum_{u\in\mathcal{N}(v)} \frac{\mathbf{W}h_u^{(k-1)}}{\sqrt{|\mathcal{N}(v)||\mathcal{N}(u)|}}\right) \tag{18}$$

where $h_v^{(k)}$ is the updated embedding of node v at layer k, $\mathbf{W}$ is a learnable weight matrix. This is similar to what happens in Fig. 4, but with a different update rule.

The update rule can be written for all nodes in a shorter form as

$$\mathbf{H}^{(k)} = \sigma(\tilde{\mathbf{A}}\mathbf{X}^{(k-1)}\mathbf{W}) \tag{19}$$

where $\mathbf{X^{(k-1)}}$ as the matrix of node features at layer $k-1$, $\tilde{\mathbf{A}}$ as the normalized adjacency matrix. Stacking multiple layers, aggregating information from further away neighbors progressively enriches each node's representation.

Spatial GCN can be enriched with attention mechanism [28,29,55] aggregating neighborhood information with attention weights. For node v, the update rule is described by

$$h_v^{(k)} = \sigma\left(\sum_{u\in\mathcal{N}(v)} \alpha_{vu}\mathbf{W}h_u^{(k-1)}\right) \tag{20}$$

where α_{vu} is the normalized (through a softmax function) attention score that can be computed as $e_{vu} = a(\mathbf{W}h_u^{(k-1)}, \mathbf{W}h_v^{(k-1)})$, with a denoting the attention function.

Another spatial graph convolution network type is the *Diffusion Convolutional Neural Network (DCNN)*. DCNNs use diffusion convolutions to capture the probability of information transmission between nodes over multiple paths [56]. The output of a DCNN layer is computed as:

$$\mathbf{Z} = \sigma(\mathbf{W} \odot \mathbf{Q}^k\mathbf{X}) \tag{21}$$

where $\mathbf{X}$ is the input matrix for DCNN, $\mathbf{Q}^k$ is a power series of $\mathbf{Q}$, the degree-normalized transition matrix of the adjacency matrix, $\mathbf{W}$ is a learnable weight matrix and $\sigma(\cdot)$ is the non-linearity (activation function). $\mathbf{Q}^*$ can be used to compute the information spreading at k-hops distance.

4.2.2. Spectral approach

Spectral GCN performs convolution operations in the Fourier domain using the properties of the graph Laplacian. Graph signals are transformed into the spectral domain, where filters are applied, and then transformed back to the spatial domain. By decomposing $\tilde{\mathbf{L}}$ into its eigenvalues and eigenvectors, that is, using $\tilde{\mathbf{L}} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T$, we can compute the graph Fourier transform of a signal x on the graph as $\hat{x} = \mathbf{U}^T x$ and the associated inverse Fourier transform is then $x = \mathbf{U}\hat{x}$, where $\hat{x}$ is the representation of x in the spectral domain. Given a signal x on the graph and denoting as g_θ a parametric function of the eigenvalues of $\tilde{\mathbf{L}}$, the spectral convolution operation $*$ is defined as:

$$g_\theta * x = \mathbf{U}g_\theta(\mathbf{\Lambda})\mathbf{U}^T x \tag{22}$$

where $g(\mathbf{\Lambda})$ is a spectral filter (typically a learnable function). The drawbacks of this approach are the computational costs of the matrix multiplication with U and the eigen-decomposition of L, which could not be viable for large graphs. To make spectral convolutions more computationally feasible, Kipf et al. [57] proposed a first-order approximation of the spectral filter through the Chebyshev-polynomials of the graph Laplacian. This approximation simplifies the computation by limiting the filter to a linear combination of the node's neighbors and itself, allowing the convolution operation to be defined as:

$$g_{\mathbf{\Theta}} * x \approx \sum_{k=1}^{K} \mathbf{\Theta}_k T_k(\hat{\mathbf{L}})x \tag{23}$$

where $\hat{\mathbf{L}}$ is the scaled and normalized Laplacian to ensure numerical stability, $\mathbf{\Theta}_k$ denotes the learnable parameters, and the Chebyshev polynomials T_k are of the first kind, that is:

$$\mathbf{T}_0 = \mathbf{I} \tag{24}$$

$$\mathbf{T}_1 = \hat{\mathbf{L}} \tag{25}$$

$$\mathbf{T}_{k\geq 2} = 2\hat{\mathbf{L}}\mathbf{T}_{k-1} - \mathbf{T}_{k-2} \tag{26}$$

While both $\mathbf{\Theta}_k$ and $\mathbf{W}$ are learnable parameters, they serve different purposes. $\mathbf{\Theta}_k$ are coefficients in the spectral convolution framework, controlling the contribution of each Chebyshev polynomial term, while $\mathbf{W}$ is a weight matrix in spatial convolution methods, directly transforming node features. The polynomial degree of the kernel determines how far the information propagates in one layer. Given that the choice is determined by the request that observed information reaches the unobserved nodes, k is usually related to the graph diameter.

4.3. Alternative graph-based architectures

As alternative graph-based architectures, Spatial-Temporal GNNs (STGNN) capture simultaneously spatial and temporal dependencies and are usually based on the implementation of Recurrent Units or 1D Convolutional Neural Networks. Considering the simplest case [29], temporal features can be captured using a Recurrent Neural Network (RNN) layer, whose output at time step t is

$$\mathbf{H}^{(t)} = \sigma(\mathbf{W}_i\mathbf{X}^{(t)} + \mathbf{W}_h\mathbf{H}^{(t-1)} + \mathbf{b}) \tag{27}$$

where $\sigma(\cdot)$ is an activation function, $\mathbf{H}^{(t-1)}$ is the previous embedding, $\mathbf{W}_i, \mathbf{W}_h$ are the learnable parameters and $\mathbf{b}$ the bias. The spatial features can instead be processed using GCN layers, obtaining

$$\mathbf{H}^{(t)} = \sigma(\mathrm{GCN}(\mathbf{X}^{(t)}, \mathbf{A}, \mathbf{W_i}) + \mathrm{GCN}(\mathbf{H}^{(t-1)}, \mathbf{A}, \mathbf{W_h}) + \mathbf{b}) \tag{28}$$

where GCN$(\cdot)$ is a GCN layer. Many variants combine LSTM with ChebNet [58], Diffusion Convolution with GRU [59], that use an additional RNN at edge level [60] or that use a 1D convolution for spatiotemporal modeling [61].

Another important GNN-based technique is the Graph Autoencoder (GAE), which belongs to the family of AEs unsupervised learning models that learn a compressed representation of input data by reconstructing it. In the context of graphs, the encoder maps nodes or entire graphs to embeddings, and the decoder reconstructs the original graph structure or properties [62]. This is based on the intuition that an AE can be used as a dimensionality reduction technique to learn low-dimensional node representation. Different architectures or algorithms can be used for the encoder and the decoder, such as MLP [63], CGNN and a similarity measure [64], a CGNN and MLP [65] or both LSTM networks [66].

5. Datasets

Assessing the performance of data-driven approaches faces significant challenges due to the limited availability of real-world datasets. One reason is the lack of the actual physical structure of WDNs or the lack of labeled data [67]. To the best of our knowledge, very few publicly available datasets based on real-world deployment are available, like the LeakDB [68] that provides two network topologies or the Water Distribution System Research Database [69] that provides 40 different topologies. These databases offer valuable datasets since they reflect real-world deployments. Yet, the size of the topology and the measurements included, as depicted in Table 2 that provides for all the benchmark datasets used by the techniques and methods surveyed, significantly limit the use of these datasets.

Very few studies utilize the topologies of pilot deployments and real-world WDN infrastructure along with data collected from their operation to validate the performance of the proposed methods [45,54,55,84–86]. For example, the experimental validation of the model developed in [55] is based on a private large-scale WDN in The Netherlands in the area of Oosterbeek comprised of 5855 junctions and 6188 pipes. The data used in these studies, the topologies of the WDN, and the measurement data are not available to the public, mainly for privacy and safety reasons.

Table 2
Features of benchmark WDN datasets used by the techniques and results surveyed.

WDN	Ref	Fig.	Junctions	Pipes	Reservoirs	Tanks	Pumps	Valves	DMA
Anytown	[70]	Fig. 6(a)	22	41	1	–	1	–	–
Hanoi	[71]	Fig. 6(b)	32	68	1	–	–	–	–
BakRyan	[72]	Fig. 6(c)	36	58	1	–	–	–	–
Fossolo	[69]	Fig. 6(d)	37	116	1	–	–	–	–
TwoLoops	[73]	–	51	65	1	–	2	–	4
Pescara	[69]	Fig. 6(e)	71	198	3	–	–	–	–
Net3	[74]	Fig. 2	91	121	2	3	3	–	–
Zhi Jiang	[69]	Fig. 6(f)	114	328	1	–	–	–	–
Modena	[75]	Fig. 6(g)	267	317	4	–	–	–	–
Rural	[76]	Fig. 6(h)	381	476	2	–	–	–	–
C-Town	[77]	Fig. 6(i)	388	429	1	7	11	4	5
D-Town	[78]	–	399	443	1	7	11	5	5
Balerma	[79]	Fig. 6(j)	443	454	1	–	–	–	–
Ky13	[69]	–	775	940	2	5	–	–	–
L-Town	[80]	Fig. 6(k)	782	905	2	1	1	3	3
Richmond	[81]	Fig. 6(l)	865	949	1	6	7	–	–
KL	[79]	Fig. 6(m)	936	1274	1	–	–	–	–
Wolf-Cordera	[82]	Fig. 6(n)	1782	1985	4	6	2	4	–
Large	[83]	–	3557	4012	–	–	–	–	–

The lack of large-scale datasets has led to the creation of synthetic datasets using numerical approaches to calculate some reliability measures connected to a WDN analytically [87]. EPANET (US Environmental Protection Agency water NETwork) is an open-source software to model hydraulic and quality dynamics of a WDN [41]. During the last 20 years, EPANET has been established as the de facto standard tool for industry and academia. The vast majority of the data-driven methods rely on EPANET to analyze the WDN using computer simulation models [43,56,88–90].

The availability of various WDN datasets with different topologies in terms of graph diameter and average node degree is essential since some GNNs suffer underreaching and/or over-smoothing. In some cases, long-term dependencies are a problem to facilitate learning. In others, wherein each layer of information is exchanged with one-hop neighbors, the depth must be too deep to capture information from further away neighbors. In these topologies, GNNs may suffer as their performance does not necessarily increase with depth. Gated solutions may be helpful in these cases.

5.1. Modelling water distribution networks using EPANET

EPANET [41] models a WDN as a collection of links connected to nodes: *links* represent pipes, pumps, and control valves; and *nodes* represent junctions, tanks, and reservoirs, see Section 2. EPANET can perform *Extended-Period Simulation* accounting for temporal variations in tank elevation conditions or demands within discrete time intervals. The flow within the network is assumed steady during each period, with tank conditions being the unique variables carried over time steps, updated by conservation of mass relationship. *Dynamic modeling* addresses the fluctuations in flow conditions, which may differ simultaneously and between network points. *Transient analysis* is used to evaluate rapid changes in flow, such as sudden valve closures or pump activations. Gradually varying conditions presuppose a rigid pipe structure, where flow changes occur instantaneously along the pipe, resulting in uniform but potentially time-varying velocities. The observed properties of all the simulated elements of the WDN are recorded for each simulated time interval and exported as a complete trace.

5.2. Sythentic datasets

Several WDN topologies have been introduced, allowing us to benchmark the performance of a given model in terms of accuracy, scalability, and robustness. Table 2 lists a limited number of benchmark networks most commonly used, indicating the network size in terms of numbers of tanks, valves, junctions, and pumps while the corresponding topologies are depicted in Fig. 6.

Specific datasets have been used for benchmarking solutions as part of competitions like Anytown first presented for the Battle of the Network Models [70], C-Town initially created for the Battle of the Water Calibration Networks [77], D-Town based on C-Town developed for the Battle of the Water Networks II [78] and L-Town created for the Battle of the Leakage Detection and Isolation Methods BattLeDIM [80].

Some other datasets are loosely connected to actual WDN infrastructure, like Hanoi [68], Richmond that depicts a WDN owned by Yorkshire Water in the UK [81], Wolf-Cordera Ranch developed by the Colorado Spring Utility [82], Modena representing a simplified version of the WDN serving the town of Modena, Italy [75] and Net 3 modeled after the North Marin Water District (NMWD) in Novato, California [74].

6. Key research areas in GNN applications for WDN

The following section explores the most prominent research areas within WDNs, examining how GNNs have been or could be applied to address challenges in these areas, illustrated in Fig. 7. The discussion structure is as follows: we define each problem and review traditional methods to tackle it, emphasizing key studies that set the stage for adopting GNNs. After a concise literature review, we focus on one or two exemplary works in detail, providing insight into innovative solutions leveraging GNNs to overcome these challenges. For a comprehensive summary, the primary GNN-related studies in WDNs and their key characteristics are presented in Table 3.

6.1. State estimation

Problem statement. The advances in sensing technology allow the use of various types of sensors, e.g., flow and pressure; however, considering size, complexity, and budget constraints, it is impossible to monitor each location of a WDN. *State Estimation* (SE) is defined as inferring the state at all locations given measurements at a limited number of locations, where in states are included the collection of heads at junctions and flows along pipes [92].

Traditional methods. Historically, the SE problem has been formulated as an inverse optimization problem. The objective function is to find the best similarities between predictions and measurements by considering physical laws such as mass balance and energy conservation. According to a review by Tshehla et al. [102], the three main criteria used in solving such problems include the Least Squares (LS), Least Absolute Value (LAV), and Min–Max criteria. The choice of error criterion depends on the expected error type: LS works well with Gaussian noise but

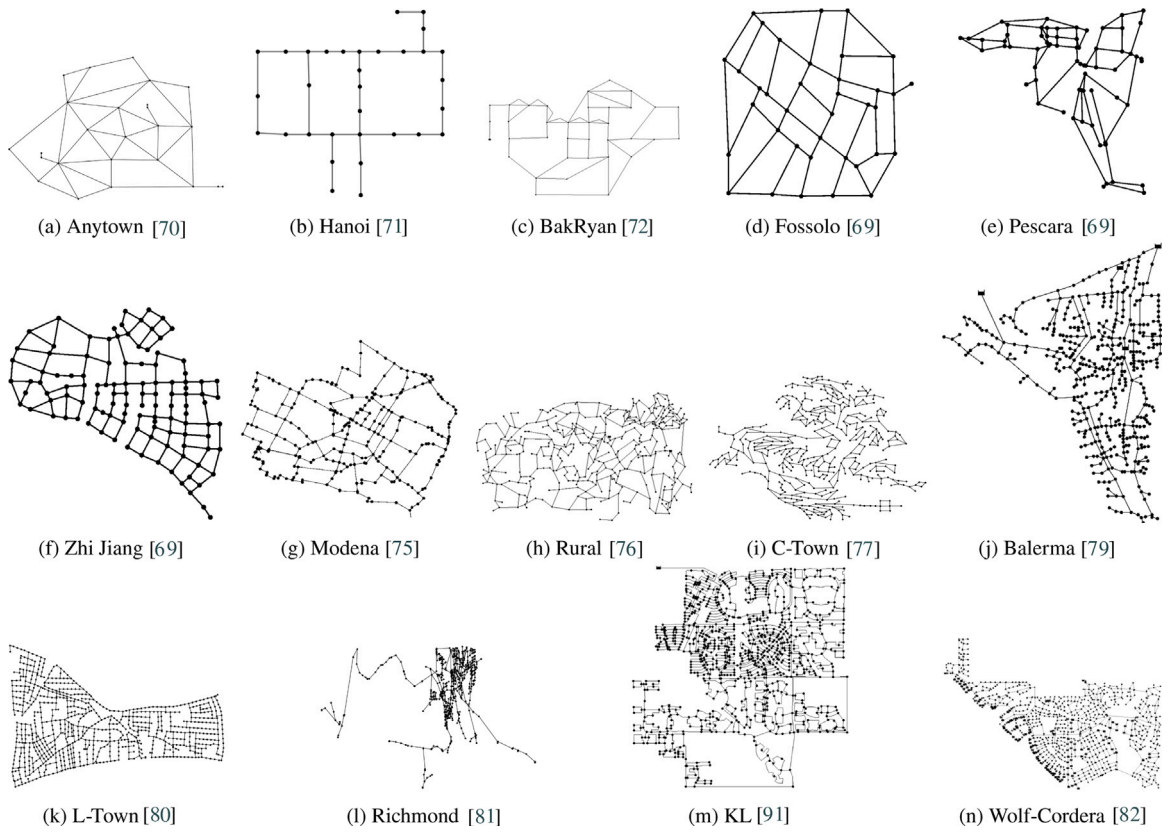


Fig. 6. Topologies of benchmark WDN datasets used by the techniques and methods surveyed [91].

Table 3
Overview of techniques and results surveyed. *Challenge (C)*: Optimal Sensor Placement (OSP, Section 6.5), Leak Detection and Localization (LDL, Section 6.2), State Estimation (SE, Section 6.1), Water Quality Monitoring (WQM, Section 6.4), Demand Forecasting (DF, Section 6.3). *Result (R)*. *Graph Abstraction Model (G)*: Graph Abstracting Junctions and Pipes (GAJP, Section 3.2), Graph Abstracting Sensing Devices (GASD, Section 3.3), Graph Abstracting District Metered Areas (GDMA, Section 3.4). *Type of GNN (GNN)*: Recurrent Graph Neural Network (RGNN, Section 4.1), Convolutional Graph Neural Network (CGNN, Section 4.2), Gated Graph Neural Network (GGNN, Section 4.1.1), Spatial GCN (GCN, Section 4.2.1), Graph Autoencoder (GAE, Section 4.3), Spatial-Temporal GNN (STGNN, Section 4.3), Spectral GCN (SpGCN, Section 4.2.2). *Evaluation Metrics (EM)*: Accuracy (Acc), top-K accuracy (Acc_K), Average Topological Distance (ATD), Coefficient of Determination (R^2), Coverage Rate (CR), Detection Time (DT), Empirical Cumulative Distribution Function of Relative Errors (ECDRE) Mean Average Error (MAE), Mean Absolute Percentage Error (MAPE), Mean Relative Absolute Error (MRAE), Mean Squared Error (MSE), Nash and Sutcliffe Coefficient of Efficiency (NSE), Normalized Root Mean Square Error (NRMSE), Ratio of positive predictions over the total number of predicted values (RPP), Relative Error (RE), Root-Mean-Square Error (RMSE), Root Mean Square Deviation (RMSD), Root-Mean-Square Logarithmic Error (RMSLE), Taylor Diagrams (TD).

C	R	G	Type of GNN	Dataset	EM
SE	[89]	GAJP	SpGCN	Anytown, C-town, Richmond	MSE, ECDRE, TRD
	[92]	GAJP	GAE	TwoLoops	R^2, RMSE
	[88]	GAJP	SpGCN	Anytown, C-Town, Richmond, L-town	MRAE
	[93]	GAJP	GCN	C-Town	NRMSE, MAPE
	[94]	GAJP	SpGCN	Hanoi, Fossolo, Pescara, L-Town, Zhi Jiang	MRAE
	[55]	GAJP	GCN with attention	Oosterbeek, Ky13, Large, C-Town, Richmond, L-Town	MAE, MAPE, NSE, RPP
	[95]	GAJP	SpGCN	BakRyan, Fossolo, Pescara, Modena, Rural, KL	R^2, RMSE, speedup
	[96]	GAJP	Graph autoencoder	BakRyan, Pescara, Net3, Zhi Jiang, Modena, L-Town, KL	R^2, MAE, RMSE
LDL	[86]	GASD	CGNN, RGNN	Modena	TPR, FPR, DT
	[97]	GAJP	GCN with pointers	L-town	ER
	[43]	GASD	SpGCN	L-Town	Euros saved, TPR
DF	[98]	GASD	GCN + Attention	Private	R^2, MAE, MAPE, RMSE, RMSLE
	[99]	GAJP	STGNN	C-Town	S score, F score
	[100]	GDMA	STGNN	Shenzhen	MAE, RMSD
	[101]	GASD	STGNN	Private	RMSD
	[85]	GDMA	RGNN	Private	R^2, MAE, MSE
WQM	[84]	GAJP	GGNN	Private	MAE, MAPE
	[56]	GAJP	GCNN and DCNN	Wolf-Cordera	Acc_K
	[54]	GAJP	GGNN	Private	TOPc
OSP	[45]	GAJP	Structural deep clustering network	Private	CR
	[90]	GAJP	SpGCN	Anytown, Net3	Acc, ATD, RMSE

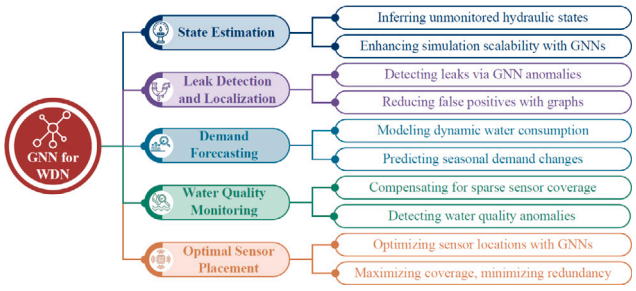


Fig. 7. Most prominent key research areas explored in GNN applications for WDN.

struggles with sparse data and large errors, while LAV is more robust to Cauchy-distributed or unknown errors. At the same time, The Min–Max criterion works well with minimal outliers. Still, both LAV and Min–Max can lead to non-differentiable, nonlinear, and sometimes non-convex problems, increasing computational complexity [92,102]. For this reason, NNs are a promising technique when data are available, reducing computational effort to the training phase.

For example, a hybrid method combining a recurrent neural network (RNN) with Newton–Raphson iterations has been proposed by Gabrys et al. [103]. In the first phase, the RNN is used alongside a numerical solver to handle the nonlinear equations describing the WDN. In the second phase, the same RNN quantifies the impact of measurement uncertainty, providing confidence intervals for state estimates by calculating bounds on the potential error. Lima et al. [104] propose using nodal pressure and inlet flow to train a simple MLP model for real-time pressure estimation for the unmonitored locations. Although the approach shows promising results, it faces several limitations, including careful selection of monitoring points, sensitivity to network changes (such as pipe substitutions or new installations), and the potential for increased uncertainty when roughness values are not consistently defined.

Graph-based methods. Xing et al. [92] identify three main challenges of using NNs for the SE problem. The first challenge is that a supervised approach requires labeled data for all samples. Furthermore, a purely data-driven approach would neglect the underlying water hydraulic laws, potentially leading to unfeasible results. A semi-supervised learning approach is needed to address these issues. Another challenge is that NNs typically operate on Euclidean data, while WDNs are represented in graph form. Thus, GNNs offer a promising solution for SE in WDNs, as they can learn from both node attributes and network topology, enabling them to capture functional relationships between monitored and unmonitored locations while respecting physical laws. Given the following parameters as inputs: topology of the WDN encoded in the adjacency matrix, demands at all junctions, heads at reservoirs, tanks, and head measurements at some junctions, and loss coefficients at all pipes, the SE goal would be to estimate all states of the WDN and the flows and the heads at each junction.

Several attempts have been made to tackle the SE problem using topological information on WDNs. For example, to evaluate the performance of GNNs for SE, Xing et al. [92] compare two learning methods: a supervised scheme trained using complete information provided by EPANET and a semi-supervised scheme trained with limited monitored locations, integrating physical laws of mass and energy conservation. Hajgato et al. [89] propose using spectral graph convolutional networks (GCNs) for pressure estimation at unknown nodes, but their method did not fully exploit the available structural information [88]. To address this, Ashraf et al. [88] enhance the GCN model by incorporating both node and edge features, tackling the challenge of long-range dependencies in sparse sensor networks. They further improved scalability with a multi-hop approach that aggregates messages over multiple layers, reducing the number of parameters while capturing information from distant neighbors. Building on these advancements, Salem et al. [93] introduce a framework for water quality state estimation (WQSE) using GNNs to predict chlorine concentrations across WDNs with limited sensors, developing both a Static Prediction (SP) model and a Dynamic Prediction (DP) model. While the SP model outperformed the DP model in accuracy and noise robustness, the DP model demonstrated flexibility for sensor placement optimization. Ashraf et al. [94] propose a novel, physics-informed DL model for hydraulic SE in WDNs, using a few GCN layers and a message-passing algorithm to infer hydraulic state features in an unsupervised manner, achieving faster emulation times and high accuracy compared to traditional EPANET simulations. Truong et al. [55] further validate the capabilities of GNNs for SE, proposing a GAT model with residual connections trained on random sensor placements and various noise levels, confirming GNN's adaptability and robustness in real-world scenarios. Kerimov et al. [95] investigated using GNNs as metamodels to estimate nodal pressures in steady-state EPANET simulations. Their results indicate that GNNs perform similarly to traditional Multi-Layer Perceptrons but take longer to execute, with preliminary findings suggesting that GNNs can learn shared representations across different networks. However, further work is needed to assess the feasibility of genuinely general metamodels. In their other work [96], they introduce Edge-Based Graph Neural Networks (Edge-GNNs) to overcome limitations in GNN transferability. By representing pipe-level processes in greater detail, Edge-GNNs improve the accuracy of pipe flow rate and pressure predictions. Their results demonstrate that Edge-GNNs outperform traditional node-based models, with strong generalization performance on unseen systems.

Case study: Reconstructing nodal pressures with spectral GNN [89]. A characteristic work in the field of SE for WDN is the one by Hajgato et al. [105] that tested the effectiveness of the GNN approach for nodal pressure recovery. The pipe network is modeled as a connected and undirected graph, where nodes represent junctions and edges represent pipes (see Section 3.2). The adjacency matrix **A** encodes node connections. While the authors initially explored the potential of incorporating hydraulic properties by weighting the edges, they ultimately found that this approach led to suboptimal results. Therefore, they opted for a binary adjacency matrix. The task is reconstructing unobserved nodal pressures based on partial graph signals, assuming fixed sensor placement. The authors used a static approach, focusing on steady-state pressure distributions and leveraging the Markov property to simplify reconstruction without modeling temporal dynamics.

Considering that the WDN topology is static, input and output variables are nodal properties without edge inference, and connection strength derives from hydraulic properties, they opted for the K-localized spectral graph filtering method ChebNet (see Section 4.2.2). Unlike other GNNs, ChebNet uses Chebyshev polynomials to efficiently learn localized filters, allowing controlled information propagation across the graph and capturing local node relationships crucial for nodal pressure prediction. This method avoids the complexity of global learning and is well-suited for modeling local hydraulic interactions.

The evaluation revealed that GNNs based on K-localized spectral filtering were effective in reconstructing nodal pressures in WDNs, achieving a relative error of less than 5% with at least 5% observation ratio on three benchmark networks: *Anytown*, *C-Town* and *Richmond*. The performance improved with higher observation ratios and outperformed traditional methods like mean imputation, which approximates missing nodal pressures using the mean of known pressures, and interpolated regularization, a Tikhonov regularization variant [106] that estimates pressures through interpolation. The researchers found that interpolated regularization maintained standard deviation but failed to accurately match ground truth values, indicating a potential flaw in the nodal pressure distribution smoothness assumption. On the contrary, their data-driven approach avoided inaccurate assumptions and outperformed other methods across various observation ratios. However, it struggled in Anytown due to the limited number of nodes. Finally, the authors noted that while most nodes displayed low relative errors, a few exhibited errors as high as 100%.

Case study: Pressure reconstruction using spatial GCN [88]. Ashraf et al. present a spatial graph convolutional network (GCN) approach to address critical challenges in water distribution systems, particularly the difficulties of missing data from sparse sensor deployments and capturing long-range dependencies in large networks. Their methodology models water distribution systems as graphs where junctions represent nodes and pipes form edges while introducing key innovations to improve pressure estimation accuracy. The approach begins by embedding node features (pressure measurements) and edge features (pipe characteristics) through separate linear transformations. A distinctive aspect of their model is the dynamic updating of edge features by incorporating absolute differences between connected node embeddings, which better captures the physical relationships in water networks. The authors developed a multi-hop architecture that reduces parameter requirements while maintaining performance across differently sized networks to enhance scalability. The complete node update process combines information from neighboring nodes and updated edge features through an MLP with SeLU activation.

The evaluation demonstrated significant improvements over existing approaches, particularly the ChebNet benchmark from [89], across four different water networks (Anytown, C-Town, Richmond, and L-Town) under various sensor density scenarios (5%, 10%, and 20% coverage). The model showed particular strength in larger networks and low-sensor conditions using mean relative absolute error as the primary metric. In the L-Town case study, incorporating pressure-reducing valve masks as additional edge features further enhanced performance, achieving reliable results with only 4.22% sensor coverage. These advancements provide water utilities with more accurate pressure reconstruction capabilities and better generalization across different network sizes and sensor configurations, offering a practical solution for real-world implementation where comprehensive sensor coverage remains economically challenging.

6.2. Leak detection and localization

Problem statement. An essential focus of optimizing and digitizing WDN research involves the detection of leaks to reduce water losses. Leaks are because WDNs are generally old, worn, and continually deteriorating [107], resulting in massive amounts of water lost due to leakages [108]. Leaks also raise concerns about safe drinking water due to potential contamination of cracked pipes, or as they might damage public infrastructure, such as roads [39]. Water leaks can be divided into *reported*, *unreported*, and *background losses* [109]. In general, reported leakages can be easily verified, therefore ensuring their repair. However, unreported or background losses can persist for much longer, leading to larger water losses. Background losses are caused by the gradual deterioration of WDN's components, such as pipes, valves, and fire hydrants, and it is assumed that they are distributed along the WDN. In contrast, unreported losses include component failure, such as pipe bursts due to a significant crack in the pipe, leading to a substantial outflow of water. Identifying unreported leaks is crucial to reducing water loss based on minimizing the duration of leaks [110].

Traditional methods. Traditional approaches to leak detection can be broadly categorized into *passive* and *active* methods [38]. Passive methods, which rely on manual inspection and acoustic listening devices, are labor-intensive and time-consuming, making them unsuitable for large-scale or real-time leak detection [108]. This has led to the development of active methods that utilize sensor data and computational models to automate leak detection. Active methods are further divided into *model-based* and *data-driven* approaches. Model-based methods rely on detailed hydraulic models to simulate normal operating conditions of WDNs. Leaks are identified as deviations from expected behavior by comparing simulated data (e.g., pressure or flow) to real-time measurements. While effective in controlled environments, these methods suffer from several limitations:

- They require extensive domain knowledge and substantial computational resources to build and calibrate hydraulic models.
- Uncertainties in model parameters, such as pipe conditions and demand patterns, reduce accuracy.
- Changes in network topology demand frequent recalibration, limiting scalability and adaptability to real-world, dynamic scenarios.

These limitations have spurred the rise of data-driven methods, which bypass the need for detailed hydraulic models by directly analyzing sensor data. Data-driven techniques leverage patterns in flow, pressure, or other system metrics to detect anomalies that indicate leaks. The effectiveness of these methods depends on the quality and quantity of available data. Wu et al. [111] demonstrate that prioritizing model selection is not always the most critical factor in achieving optimal performance. Instead, they show that employing data augmentation techniques such as jittering, scaling, warping, and Iterated Amplitude Adjusted Fourier Transform can significantly enhance the performance of leak detection methods. These results suggest that carefully designed preprocessing and augmentation strategies can play a pivotal role in improving model robustness and generalization, sometimes outweighing the impact of more complex or sophisticated model architectures. Wu et al. [37] categorize data-driven methods into *classification*, *prediction-classification*, and *statistical methods*. *Classification methods* involve distinguishing data affected by bursts from normal data. For instance, Garlisi et al. [112,113] proposes a framework that identifies leaks through a supervised classification approach using nodal features such as demand, head, and pressure. Ravichandran et al. [114] introduce a multi-strategy ensemble learning approach that utilizes a gradient-boosting tree classification model for detecting leaks based on acoustic signals. Acoustic signals have also been used to obtain scalogram images for leak diagnosis through a CNN [115]. Additionally, Adraoui et al. [116] use a Random Forest classifier to identify leaks by extracting features from daily pressure signal data, employing wavelet decomposition for feature extraction. *Prediction-classification methods* operate on the premise that, in the presence of an anomaly, the prediction output consistently deviates from the measured data. These methods require only normal hydraulic data to build a prediction model. However, obtaining clean data necessitates the removal of outliers and abnormal data, prompting the use of data preprocessing and statistical methods. For example, Li et al. [117] present a leakage location framework based on a Residual Network that uses parallel classification and regression to accurately pinpoint the leak location in a WDN, with a multi-supervision mechanism to improve convergence. Wang et al. [118] discuss challenges in leak detection in WDNs, including setting proper thresholds for classification, prediction bias from outliers, and the accumulation of prediction residuals, which complicate real-time applications and the continuous identification of anomalies. *Statistical methods* for burst identification do not rely on prediction or classification models. However, in most cases, detection hinges on statistical theory, often falling under statistical process control. The latter entails monitoring process variation caused by anomaly events through control charts and analytic tools. Although not requiring complex data-driven models, statistical methods may be subject to high uncertainty. Statistical methods have also been combined with deep learning. Ahmad et al. [119] proposes to extract features from acoustic emission signals through a novel vulnerability index derived using a multiscale Mann–Whitney test, using a 1D CNN for leak detection and size identification.

Beyond purely data-driven approaches, hybrid methods integrate data-driven leak detection with model-based leak localization, leveraging sensor data patterns and hydraulic simulations for improved accuracy. For example, Daniel et al. [120] follow a hybrid approach by identifying leaks using semi-supervised linear regression on pressure data and refining leak localization through a simulation-based optimization framework.

Graph-based methods. The rise of graph-structured approaches has addressed data-driven challenges, offering closer representations of problems in non-Euclidean spaces. Shekofteh et al. [121] propose a method for leak detection in WDNs by using graph theory to decompose the network and an artificial neural network to identify leakage areas based on pressure data and demand uncertainty. Barros et al. [15] use graph signal processing tools and centrality analysis (specifically PageRank) to detect leaks based on changes in pressure signals, leveraging statistical correlation to identify and localize leaks. The research on GNN for leak detection and localization is currently limited. Gardarsson et al. [43] combine data-driven techniques and model-based logic to detect leaks by using GNNs (in particular, Chebyshev spectral graph convolutional network) to predict and reconstruct nodal pressures in a WDN. By analyzing the residual signal derived from the discrepancy between predicted and reconstructed pressures, they detect leaks and localize them by leveraging the topological information embedded in the GNN models. Zhang et al. [97] highlight that data-driven methods often struggle to generalize to out-of-distribution data, and to address this limitation, they propose an algorithm-informed graph neural network (AIGNN) for leakage detection and localization in WDNs, inspired by the perfect generalization ability of classical algorithms. Building on Gardarsson's work [43], AIGNN is trained to emulate the Ford-Fulkerson algorithm for max-flow problems, with two models deployed to reconstruct and predict pressures based on current and previous measurements, respectively, detecting leaks by analyzing discrepancies between the outputs.

Case study: GNN for leak detection [86]. In this work, Zanfei et al. propose a classification model to identify bursts from abnormal data in a dataset of pressure and flow measurements. The monitored locations define the graph nodes, and their features are composed of pressure or flow values depending on the type of sensor. The adjacency matrix is built based on the correlation coefficient between all the time series, following the correlation-based graph model approach described in Section 3.3. Two models are proposed.

GCN Model (Section 4.2): Designed to extract spatial information from the data, it is composed of three layers of GCN and followed by three dense layers. The final output is a binary classification based on a threshold applied to match the burst/no-burst condition.

GCRNN Model (Section 4.1): This model processes both spatial and temporal information using a feature matrix with five hours of past data. The architecture includes three GCN layers to extract spatial information and two Long Short-Term Memory (LSTM) layers to process the temporal dependencies. The final output is generated through a dense layer with a threshold for binary classification.

For both models, the threshold for generating binary output was chosen to maximize detection rates while minimizing false positives on the training dataset. The methods were tested on the Modena network (Section 5). The data for the study was generated following the procedure described by Menapace et al. [122], simulating water demands with deterministic trends and normally distributed randomness. Burst events were modeled with a Poisson distribution, representing pipe crack types such as longitudinal, circumferential, and spiral. The flow rate time series were collected from the outflows of tanks and reservoirs, while pressure time series were retrieved from selected monitoring points. The selection of these points follows the method described by Righetti et al. [123], ensuring that the selected nodes have high sensitivity in detecting pressure variations from different leak positions while minimizing redundancy by avoiding highly correlated nodes.

The proposed GCN and GCRNN models were compared against two Multi-Layer Perceptrons (MLPs): MLP1, which uses pressure and flow data from the same time step, and MLP2, which incorporates five hours of past measurements, similar to the GCRNN. Evaluation metrics include True Positive Rate (TPR), which indicates the ability to detect actual bursts, Detection Time (in hours), which measures the delay in burst detection, and False Positive Rate (FPR), which counts the number of false alarms triggered by the models. The experimental results demonstrate the effectiveness of the proposed models for detecting burst events in WDN. Both models achieved nearly perfect TPR, indicating high accuracy in burst detection. The GCN model outperformed the GCRNN model in detection speed, identifying bursts almost instantly, while the GCRNN model showed more distributed detection over time. The GCN model had fewer false positives, mostly lasting one hour, while GCRNN produced more under high demand variability. Both outperformed MLP models, which had lower TPR and longer detection times. Both models showed greater robustness across different conditions, including high seasonality.

Case study: Leak detection via reconstruction and prediction [43]. Gardharsson et al. propose a GNN-based method for leak detection and localization that augments a physics-based hydraulic model with historical data from selected pressure monitoring points. The WDN topology is a graph of junctions and pipes, as described in Section 3.2.

The adjacency matrix $\mathbf{A}$, capturing node connectivity, is constructed using three weighting schemes:

- *unweighted edges*: $\mathbf{A}_{uv} = 1$;
- *pipe length weighted*: $\mathbf{A}_{uv} = l_{uv}$ with l_{uv} being the pipe length;
- *hydraulic loss weighted*: $\mathbf{A}_{uv} = 4.727 \cdot C_{vu}^{1.852} \cdot \frac{d_{vu}^{4.871}}{l_{vu}}$ with reference to Hazen–Williams equation.

Additionally, self-loops were added to the graph to enhance the influence of pressure monitoring points in the signal reconstruction process.

The method relies on two main components: **reconstructor**, which retrieves nodal pressures for a specific instant using sparse pressure measurements, and **predictor**, which predicts future pressure measurements based on historical data. Let the observed pressure data at time t on the graph $\mathcal{G}$ be denoted as $\mathbf{x}_{\mathcal{G}}^{(t)} = [x_1^{(t)}, x_2^{(t)}, \dots, x_N^{(t)}]$, where most values are NaN, reflecting the limited number of pressure monitoring points. The reconstructor infers the missing pressures $\hat{y}_r^{(t)}$, and the predictor uses historical data over a time window T, denoted as:

$$\mathbf{X}_{\mathcal{G}}^{(t-T:t-1)} = \left[\mathbf{x}_{\mathcal{G}}^{(t-T)}, \mathbf{x}_{\mathcal{G}}^{(t-T+1)}, \dots, \mathbf{x}_{\mathcal{G}}^{(t-1)}\right],$$

to predict the pressures at the next time step $\hat{y}_p^{(t)}$.

The residual signal is defined as the difference between the predicted and reconstructed values:

$$\mathbf{r}(t) = \hat{y}_p^{(t)} - \hat{y}_r^{(t)}.$$

Under normal (non-leak) conditions, the residual signal remains stationary, influenced only by random noise. A deviation from this behavior indicates the presence of a leak. The residual signal on edges to localize the leak is defined as:

$$r_{uv}(t) = r_v(t) - r_u(t),$$

where u and v are the nodes connected by an edge, a leakage is detected when the moving average of the residual, $\bar{r}_{uv}(t) = \frac{1}{m}\sum_{i=0}^{m-1} r_{uv}(t-i)$, exceeds a threshold $\bar{\rho}_{uv}(t) = \alpha\bar{\sigma}_{uv}(t)$, where α controls the rate of false alarms and $\bar{\sigma}_{uv}(t)$ is the standard deviation over the window.

To ensure accurate localization, the authors applied a *Chebyshev spectral graph convolutional network* (Section 4.2.2), which preserves high-frequency (localized) information. For the reconstructor, the input channel size ch_{in} is set to 1, while for the predictor, it is T, corresponding to the time window size.

The methodology was validated on the *BattLeDIM* [124] dataset from the *L-Town* network. The EPANET nominal model and sensor measurements from 2018 were used to train the reconstructor and predictor, with documented leak information (size and inflow) as ground truth. The evaluation employed two primary metrics: the Economic score (Euros saved), which reflects cost savings from early leak detection by reducing non-revenue water and repair costs, and the True

Positive Rate (TPR), which measures the proportion of correctly detected leaks within a 300-meter radius of the actual leak location. The model was tested on unseen 2019 data, using a window size of 7 days and $\alpha = 1$, filtering candidates to report only one detection per neighborhood with a 6-hop radius each day. Although the proposed methodology achieved competitive results, it suffered from 138 false positives over a year. It is important to note that the model only used pressure measurements, while other teams utilized flow and smart meter data. This could partially explain the higher number of false positives. Despite this limitation, the method demonstrated robustness in localizing leaks based on pressure data alone.

6.3. Demand forecasting

Problem statement. Accurate urban water demand prediction is essential for meeting residents' needs and planning infrastructure expansion. Estimations under climate and population growth scenarios guide strategic development. Reliable predictions ensure cost-effective sizing and operation of reservoirs, pumps, and pipes. The initial step involves developing accurate models for forecasting water demand, mainly focusing on assessing peak demand. Two categories of demand forecasting exist: *short-term* forecasting, utilized for operational purposes, and *long-term* forecasting, crucial for planning and infrastructure design [125]. Forecasting methods, which range from traditional statistical techniques to advanced ML models, vary in complexity based on spatial and temporal granularity, and the choice of method depends on factors such as forecasting horizon, data availability, and system complexity. ML methods, time series, and pattern-based approaches are typically employed for short-term forecasting, while econometric models are more commonly applied for long-term predictions [126].

Traditional methods. Traditional statistical techniques, such as time series analysis (e.g., ARIMA models), are commonly used [127–130] and rely on historical data patterns, assuming stationarity in trends and seasonality. While effective for stable demand patterns, they may struggle with abrupt changes in consumer behavior or external factors [131]. Conversely, modern water demand forecasting increasingly relies on ML models, which can capture nonlinear relationships and interactions between various influencing factors [132]. Incorporating external variables like weather data, economic indicators, and demographic trends can also significantly affect forecasting accuracy [128,133]. Commonly used ML methods include regression models (e.g., Random Forest, Gradient Boosting) [134] for handling datasets with numerous features like weather conditions, population growth, and economic indicators, and DL models, such as LSTM [131,134–137], which excel in processing sequential data and capturing complex temporal dependencies.

Graph-based methods. Spatial forecasting approaches, such as spatial–temporal models, consider the geographic distribution of demand. These methods, often powered by GNNs, model water networks as graphs where nodes represent demand zones, enabling more precise forecasting for localized areas. For example, Tsiami et al. [99] propose to train a temporal graph convolutional network to predict the current SCADA measurements and later utilize these predictions to detect cyber–physical attacks on WDNs by comparing them with the actual observed values. Liu et al. [98] propose a hybrid model for short-term urban water demand prediction that integrates spatial and temporal characteristics. The model combines anomaly detection using the Prophet model, spatial dependency extraction with MIC and GCN, and feature enhancement with a multi-head attention mechanism, with predictions generated by a three-layer LSTM network. Jia et al. [100] propose a Dynamic Graph Generation (DG) module using attention mechanisms to capture global dependencies and local dynamic changes in water demand across neighborhoods alongside a Spatial-Temporal Feature Network (STFN) combining graph convolutional networks and 1D-CNNs for water demand forecasting. Wu et al. [101] identify the limitation of current GCN techniques in assuming equal sensor importance and purely data-driven spatial correlations. To address this, they propose a spatial correlation-based GCN (SCGCN) model using a self-attention mechanism to analyze water demand data and flow resistance (head loss), generating hydraulics-informed spatial correlation coefficients. These coefficients weight neighboring sensors' data for accurate feature extraction in the SCGCN model.

Case study: GNN for demand forecasting [85]. Zanfei et al. present a methodology for short-term forecasting of water demands. Starting from the consideration that the demands of users in geographically close areas exhibit a strong correlation, this solution represents the demand time series of neighboring WDNs with a graph abstraction model similar to the one described in Section 3.3. In this case, however, nodes represent *aggregated water demand across DMAs*, rather than individual sensor points within the network, while edges capture correlations in demand between different DMAs. The solution frames the graph as a spatio-temporal graph, where each node feature matrix has dimensions $N \times T$, and node features are updated dynamically over time. The model aims to predict the subsequent water demand for each DMA (node) based on the aggregated past demand values. The authors combine graph convolution operations (Section 4.2) with recurrent units (Section 4.1) to effectively handle the water demand forecasting problem. The GCN layers extract spatial information from the data, and the output is given to the LSTM layers to capture temporal features. The final dense layer is used to provide the final prediction.

The evaluation uses 50 water demand time-series data over seven years, with hourly resolution, from small-scale WDNs (up to 10,000 users) within 30 km of each other. The first six years were used for training and validation, with the last year as a test set. Four graphs were created based on correlation coefficients: *Graph 1* used strongly correlated time series with shared physical connections, *Graph 2* included time series with at least 0.7 correlation, *Graph 3* used at least 0.6 correlation, and *Graph 4* included those with at least 0.5 correlation. Optimal GCRNN architectures were selected for each graph after the grid search.

Graph 1 performed best, indicating that more straightforward graphs with less time series may lead to better predictions. This graph structure was then compared with two other approaches: SARIMA (Seasonal Autoregressive Integrated Moving Average) and a state-of-the-art LSTM model. The GCRNN model consistently outperformed the LSTM and SARIMA across all-time series, demonstrating its superior forecasting capability. However, it was computationally more expensive, taking five times longer to train than LSTM.

The evaluation also considered a scenario where some sensors malfunctioned, resulting in missing data. For this test, two 14-day periods of sensor malfunction were simulated for three of the six-time series in *Graph 1*. Since the model predicts hourly water demand, missing values were substituted with the predicted values. While LSTM slightly outperformed the GCRNN in the first few days after the malfunction, the GCRNN maintained better performance over time, thanks to the graph structure capturing the relationships between the time series, which allowed it to make more reliable predictions despite missing data.

6.4. Water quality monitoring

Problem statement. Ensuring the safety and quality of drinking water is a critical concern for water systems management. Water quality monitoring plays a vital role in detecting and mitigating potential risks. Recent advancements in digital technologies have introduced online water quality sensors as a promising solution for real-time tracking in WDNs. However, the limited availability of sensors poses a challenge, making it difficult to monitor water quality across entire networks comprehensively. Data serves not only for ongoing water quality monitoring but also for identifying contamination sources when detected. *Contamination source identification* (CSI) is a challenging problem due to several reasons [54,56].

Traditional methods. According to Zhou et al. [56], CSI methods can be grouped into four categories: inverse problem approaches, probabilistic modeling, simulation optimization, and AI-based techniques. The inverse problem approach [138,139] relies on hydraulic and water quality models to backtrack contamination sources but struggles with dynamic demand patterns that introduce uncertainty [140]. Probabilistic models such as Bayesian networks [140,141] incorporate uncertainty and prior knowledge, allowing dynamic contamination estimation. Simulation–optimization methods [142–145] integrate hydraulic simulations with optimization techniques, achieving robust but computationally expensive solutions. Meanwhile, deep learning models leverage large datasets for data-driven CSI, offering efficiency but requiring high-quality training data to generalize well [146,147].

Graph-based methods. Each category offers distinct strengths and limitations. Traditional methods like simulation optimization provide accuracy but are computationally intensive, unsuitable for real-time detection, and often restricted to simple scenarios with single contamination sources. While scalable and fast with real-time sensor data, DL methods demand extensive high-quality datasets and are susceptible to overfitting, which can propagate model errors into CSI algorithms and compromise outcomes [145]. Moreover, traditional neural networks struggle to capture dynamics at unmonitored nodes, underscoring the need for innovative solutions.

Graph-based learning, particularly GNNs, presents a promising solution by leveraging network topology and sensor measurements to improve water quality prediction and contamination source identification. Although research in this area remains limited, existing studies show promising results. For example, Li et al. [54] propose a gated graph neural network (GGNN) incorporating hydraulic flow directions and water quality data, using a masking operation during training to enhance accuracy. Similarly, Li et al. [84] propose a GGNN for predicting chlorine concentration across a WDN using limited sensor data. Their approach integrates network topology and historical water quality measurements into a GGNN model, leveraging a masking operation during training to enhance generalization to unseen nodes. The model is evaluated on the Yantian network in China, demonstrating improved accuracy influenced by sensor placement strategies. The results highlight the effectiveness of GGNNs in capturing water quality dynamics and the impact of optimized sensor positioning on prediction performance. Zhou et al. [56] tackle the limitation of focusing on a single WDN by introducing a GCN-based framework capable of transferring knowledge from historical contamination events in one network to another, enabling cross-system generalization. They employ graph segmentation to extract local network structures and align sensor data across systems. The model learns contamination patterns from a source WDN and applies them to a target WDN, demonstrating effective transferability despite structural differences. Results indicate that leveraging shared local characteristics enhances CSI accuracy, particularly in newly built or data-scarce WDNs.

Case study: Real-time water quality prediction with GGNN [84]. In water quality monitoring, an important indicator is chlorine concentration, a disinfectant that ensures microbiological safety. Li et al. propose a Gated Graph Neural Network, GGNN, to predict water quality in terms of chlorine concentration in all nodes of a WDN, starting from a few sensor measurements. They addressed the generalization ability of their model to the unique structure of the GNN they proposed and to the use of a mask operation introduced in training to improve the learning of water quality values at unseen nodes. The GGNN methodology they proposed takes input network topology and historical chlorine concentration obtained from water quality sensors. These are two significant steps: data processing for input to the GNN and model construction.

Assume a WDN comprises n nodes, m pipes, and N_s water quality sensors. The pipe network with sensors is modeled as a connected and undirected graph, where nodes represent junctions and edges represent pipes (see Section 3.2). Flow directions, obtained from EPANET, are used to build the adjacency matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ for a directed graph with $\mathbf{A}_{ij} = 1$ when water flows from node i to node j. No weights, apart from directions, are assigned to edges. Historical water quality data are obtained using Supervisory Control and Data Acquisition, SCADA, with time length denoted by T_c. These data are then used as attributes for monitored nodes, while 0 s are used for unmonitored attribute nodes. Node features can be represented by $X \in \mathbb{R}^{n\times N_c}$ where N_c is the number of water quality measurements during period T_c.

For considering bidirectional information of flow, the adjacency matrix is expanded in $\bar{\mathbf{A}} = [\mathbf{A}, \mathbf{A}^T] \in \mathbb{R}^{n\times 2n}$. Node attributes are mapped in the initial hidden state $h_v^{(0)}$ with a mapping function from $\mathbb{R}^{N_c}$ to $\mathbb{R}^M$, where M is a hyperparameter. Starting from $\bar{\mathbf{A}}$ and $h^{(0)}$, GGNN recursively compute, similarly to what is done in [54], node states for K steps until obtaining $h^{(K)} \in \mathbb{R}^{n\times M}$, using aggregation and propagation with GRU mechanism. Higher values of K increase memory requirements and slow the training, while lower values of K limit the number of learnable dependencies between nodes. Finally, the output of GGNN is fed in a linear layer transforming $h^{(K)}$ in $\hat{Y} \in \mathbb{R}^n$ representing next time step predicted states, chlorine concentrations, for each node.

For these reasons, they replaced input features of 20% of nodes with zeros in each training batch, including all sensors for loss computation, considering measurements and predictions. Thus, masking is a regularization technique that simulates the unavailability of no-sensor node data.

The evaluation is based on data from a real-world WDN in China called the Yantian network using MAE and Mean Absolute Percentage Error. EPANET has been used in water quality simulations, where sensors have been placed using Girvan–Newman clustering based on edge-betweenness centrality and the selection of sensor nodes in each cluster. The selection process uses two methods: the betweenness centrality method, based on the idea that higher scores lead to more efficient flow and information transmission nodes, and the analysis of variance, ANOVA, based on the collection of chlorine concentration data for each node in a cluster for 5 days and the analysis of p-value to identify the most significant difference among a node and the others as the optimal one.

They also experimented with the impact of the masking operation on model accuracy by doing training and evaluation with different masking percentages and the impact of different sensor placement strategies on prediction accuracy, which showed how predictions are improved by choosing a sensor placement method based on water quality information.

Case study: Cross-network CSI [56]. Existing CSI methods focus on a single WDN without enabling knowledge transfer to other systems. This limitation is critical for new WDNs with little contamination data. Given the intuition that different WDNs share local characteristics, Zhou et al. proposed GrassL, a GCN-based approach that transfers knowledge from historical contamination events across WDNs.

The WDN is represented using the graph abstraction of junctions and pipes (Section 3.2). Contamination events are modeled as sensor data $\mathbf{X}$ and contamination source nodes y. Given a source WDN $\mathcal{W}$ and a target WDN $\bar{\mathcal{W}}$, the model learns a function α that leverages topological and historical contamination data to identify contamination sources in the target domain by minimizing:

$$\sum_{1\le i\le B} \mathcal{L}(\alpha(\bar{\mathbf{X}}_i, \bar{\mathbf{A}}), \bar{y}_i).$$

For learning, GrassL employs CNNs for temporal dynamics and Diffusion Convolutional Neural Networks (DCNNs) for spatial relationships (Section 4.2.1). A fully connected softmax layer predicts the contamination source by selecting the node with the highest probability.

Using EPANET [41], the authors split the *Wolf-Cordera* model into source (1494 nodes) and target (292 nodes) domains, simulating 10,000 contamination events. GrassL outperformed five baselines, including

CNN-DNN, LSTM-DNN, and CNN-s-GCN, in top-K accuracy (Acc_K), a measure of how well the model identifies the source of contamination among the top predictions. The C-DCNN model, a key component of GrassL, consistently outperformed all other methods, showcasing its ability to capture spatial and temporal information effectively. Compared to traditional approaches like CNN-DNN and LSTM-DNN, C-DCNN demonstrated substantial improvements in accuracy, emphasizing the strength of graph-based convolution for spatial data and CNNs for temporal features. Compared to more advanced methods, such as LSTM-DCNN and CNN-s-GCN, C-DCNN maintained a significant lead, benefiting from its ability to integrate multi-level connections and handle temporal data more consistently. Overall, the results confirm that GrassL, powered by C-DCNN, offers superior performance and provides a robust method for transferring CSI knowledge across different WDNs.

6.5. Optimal sensor placement

Problem statement. Nearly all water utilities use physics-based numerical models to simulate and analyze their WDN [148]. These models plan expansions and modifications, investigate operational issues, schedule maintenance, and estimate operating costs. Although the starting point of the numerical analysis is based on field measurements and construction drawings, it is impractical to conduct these flow tests on every pipe in the network [149] due to resource constraints. Similarly, deploying embedding wireless sensing devices across the entire WDN to provide the required flow rate data to calibrate the numerical models is unrealistic due to its size and complexity, as well as the budget and resource constraints faced by water utilities [150]. Thus, the problem of identifying the *optimal sensor placement (OSP)* constitutes a critical challenge, demanding meticulous consideration due to its direct impact on the system's scalability in terms of efficiency, reliability, and safety.

Traditional methods. Previous research has focused on two primary scenarios: placement for daily management and operation optimization and placement for detecting abnormal behaviors like leakages or contamination, often inferred from turbulence in monitored measurements. Approaches to OSP fall into three main categories: *empirical*, *optimization-based*, and *topological* methods, each offering unique strengths and limitations.

Empirical methods, commonly used in practice, rely on engineers' experience and educated guesses, often leading to suboptimal sensor placement. Network topology, hydraulic conditions, and vulnerability assessments must be considered. Still, when the selected locations are less than ideal, the collected data may fail to provide sufficient information for accurate system calibration, forcing the engineer to choose between collecting more data at additional expense or using a less precise model [149].

Optimisation-based methods frame the OSP problem as single- or multi-objective tasks, employing techniques such as integer programming, heuristic algorithms, genetic algorithms, and graph-theoretic approaches [151]. For instance, Rayaroth et al. [152] introduce the RDTBC-SFLO technique, combining a Random Decision Tree Bagging Classifier to analyze pressure data with Shuffled Frog Leaping Optimization to enhance sensor placement accuracy for water leakage detection. Hu et al. [153] propose a modified NSGA-III algorithm that addresses the NP-hardness of OSP by optimizing objectives such as minimizing detection time and maximizing detection likelihood. Similarly, Hu et al. [154] improve hierarchical algorithms by integrating joint entropy with single-sensor failure scenarios, ensuring better robustness against sensor malfunctions. Many optimization methods primarily focus on hydraulic features, neglecting the importance of nodal connectivity and its influence on sensor placement. Additionally, they typically target specific operating conditions despite the variability in hydraulic dynamics caused by changes in system operations and consumer demand.

Graph-based methods. Topological methods utilize structural properties of the network, avoiding reliance on hydraulic modeling. For example, Peng et al. [45] propose utilizing the topological characteristics to first divide a network into monitoring partitions using the Structural Deep Clustering Network algorithm before identifying burst locations using an indicator tensor to calculate hydraulic characteristics and selecting nodes with the highest partition perception rate for each partition. Cheng et al. [155] propose a feature selection algorithm leveraging only structural and connectivity characteristics and GSP to optimize sensor placement in WDNs. Their method iteratively analyzes node importance and redundancy, using graph signal reconstruction to recover pressure data and evaluate the trade-off between the number of sensors and reconstruction error. Salem et al. [156] combine a Dynamic Prediction Graph Neural Network (DP-GNN) with a genetic algorithm (GA) to optimize water quality sensor placement, leveraging the DP-GNN for chlorine concentration prediction and the GA for maximizing state estimation accuracy. Li et al. [84] convert pressure data into a band-limited signal with a graph Fourier operator and use a noise-robust sampling method to select representative nodes for sensor placement.

Case study: Optimal sensor placement for leak localization [90]. One of the tasks of OSP is leak detection. Optimal Sensor Placement (OSP) for leak detection in WDNs is crucial for minimizing water loss, structural damage, and costs. However, irregular conditions like complex topologies, fluctuating demand, and noisy data make OSP highly challenging. Traditional unsupervised approaches, based on assumptions of normal network behavior, often fail in real-world scenarios where the occurrence and nature of leaks are unknown. To overcome these limitations, Zhang et al. reformulated OSP as a semi-supervised graph node classification problem, using graph-based ML to address these irregularities effectively.

The methodology models the WDN as an undirected graph $\mathcal{G} = (V, E)$ where nodes represent junctions and edges represent pipes (see Section 3.2). A transient model simulates how leaks affect pressure data under dynamic conditions, capturing the evolution of pressure and flow rates over time. The authors employed the WNTRSimulator from the WNTR [157] Python library, interfacing with EPANET, to simulate leakage scenarios. The model assumes turbulent water flow and large leaks in steel pipes, with the leakage flow rate modeled by Eq. (11), quantifying water flow based on pressure and leak size.

The approach is divided into three distinct steps to address the reformulated semi-supervised OSP problem:

1. Create pseudo-labels. The unsupervised nature of the OSP problem is transformed into a semi-supervised task by generating pseudo-labels. Nodes (excluding tanks, pumps, and reservoirs) are grouped using the K-Means clustering algorithm into C clusters, where C is the desired number of sensors. Each cluster's central node is assigned a pseudo-label ranging from 0 to $C - 1$. These pseudo-labels serve as initial guidance for sensor placement optimization;
2. Division of monitoring areas. A two-layer GCN is used to identify monitoring areas within the WDN. The GCN processes the graph structure and pseudo-labels to define boundaries for regions where sensors will be placed. The authors used a GCN formulation based on spectral methods (see Section 4.2.2) to ensure nodes with similar pressure behavior are grouped;
3. Select representative nodes. Representative nodes for each monitoring area are chosen as sensor locations using a cross-correlation function. This function evaluates the similarity of pressure data between nodes within the same region. The sensor location is selected as the node with the highest cumulative cross-correlation with others in its region.

With sensors optimally placed, leak localization is performed using a Bidirectional Long Short-Term Memory (BiLSTM) model. The approach is validated on two benchmark WDN datasets: Anytown, the simpler network for initial testing, and Net3, a more complex network to assess scalability and robustness. The evaluation of the task included three key metrics — accuracy (ACC), average topological distance (ATD), and root-mean-square error (RMSE) — with the proposed method outperforming the baseline in each of these metrics for both datasets.

6.6. Alternative methods to GNN

While GNNs have shown promise in modeling WDNs, recent advances in conventional ML and statistical analysis address the previously mentioned core challenges through diverse methodological frameworks. We here summarize some a selected number of recent results to highlight the importance of using conventional ML tools.

For state estimation, Wang et al. [158] presents a probabilistic state estimation method for water networks that efficiently quantifies uncertainty in system states using hydraulic models and linear algebra, outperforming Monte Carlo simulations. Traditional deep learning (DL) remain competitive for leak detection, with novel approaches using convolutional neural networks (CNN) architectures on kriging-interpolated pressure images achieving 98% true positive rates [159], while prototype-based learning strategies like generalized relevance learning vector quantization (GRLVQ) classifiers achieve 93.98% accuracy in operational district metered areas (DMAs) [160].

Demand forecasting has seen breakthroughs through DL frameworks combining long-short term memory (LSTMs). For example, Wang et al. [161] introduces a solution that enhances short-term prediction accuracy for sustainable urban water management. It achieves mean absolute error (MAE) < 1.12 and squared error (R^2)$> 99\%$, outperforming benchmarks by adapting hyperparameters dynamically.

For water quality monitoring, Majnooni et al. [162] used Gated Residual Variable Selection (GRVS) networks which demonstrated superior performance, achieving $R^2 = 0.98$ for dissolved oxygen and $R^2 = 0.84$ for chlorophyll-a. Using minimal inputs like pH, depth, and temperature, GRVS models provided accurate predictions with interpretable insights via SHAP analysis, identifying pH as the most influential factor.

Optimal sensor placement challenges are addressed through quantum-inspired combinatorial optimization, reducing computational complexity for large networks while maintaining 95% coverage efficiency [163], complemented by genetic algorithm (GA) frameworks like GA-Sense that optimize sensor locations using time-series flow patterns and leak magnitude weighting. Hybrid methods integrating hydraulic simulations with linear regression show particular promise for leakage localization in low-data regimes, achieving 25% faster detection than pure data-driven models [160].

7. Future research and open issues

The recent advances in GNNs have driven significant progress in addressing key challenges associated with WDNs, enabling the application of techniques that effectively leverage their intrinsic graph-based structure. Still, several challenges and open issues remain; we summarize them in Fig. 8.

One future research direction is related to the modeling assumptions and the approaches followed to abstract the essential elements of WDNs using graph theory. Most of the works [43,89,90,92] model the WDNs by abstracting junction and pipes with graph nodes and edges, or abstracting sensing devices [86] as nodes. The selection of an appropriate graph abstraction for WDN modeling depends on the study's specific objectives and the constraints related to the available data, infrastructure, and computational resources. Abstracting junctions and pipes should be chosen to model the physical structure and perform simulations based on hydraulic principles. This approach is ideal when a detailed topological structure of the WDN and a complete knowledge of sensor data are available. Abstracting sensing devices should be preferred in scenarios with sparse sensor deployment or limited data availability as they simplify the WDN model by excluding unmeasured junctions. The majority of the literature focuses on pressure values. However, it is essential to note that hydraulic flows can directly be measured using flow sensors and, therefore, are easy to acquire when an IoT infrastructure is deployed to monitor the state of the WDN. Abstracting water flows is a key strategy for edge-level analysis in large-scale IoT networks, improving efficiency. Exploring alternative abstractions remains an open challenge for developing more accurate and computationally efficient GNNs. Current approaches focus on modeling districts rather than individual devices to simplify analysis and reduce complexity.

GNN complexity is a relevant open issue in different aspects. On the one hand, the computational complexity can pose limitations in scalability, especially in larger WDNs with components such as valves, pumps, and various hydraulic scenarios such as seasonal variations. Developing scalable algorithms that address high-dimensional graph data without sacrificing computational efficiency or prediction accuracy is part of this research direction. On the other hand, while GNNs offer substantial predictive capabilities, their complexity often positions them as black-box solutions. This motivates the need to focus on explainability [164], developing domain-specific strategies that align GNN learning with WDN principles to improve their interpretability and usability. One critical area involves integrating physical and hydraulic principles in the learning processes. Physics-informed neural networks [165] provide a promising avenue to incorporate domain-specific equations directly into the training phase, reducing data demands and ensuring consistency with physical laws. Similarly, constrained learning [166] can be used to enforce hydraulic constraints, such as ensuring pressure predictions remain within realistic bounds, potentially leading to more accurate and reliable predictions. Robustness and uncertainty quantification are additional critical aspects for real-world applications, and the integration of GNNs with techniques such as conformal prediction [167] can be employed to provide reliable uncertainty estimates, ensuring more robust decision-making in operational contexts. Furthermore, research should investigate scenarios where sensitive or proprietary data cannot be shared openly. Privacy-aware learning [168] presents a promising research direction to enable model development without compromising data security.

Emerging technologies like quantum computing and edge computing offer promising solutions to enhance GNN scalability for WDNs. Quantum computing could address complex optimization problems, such as sensor placement or large-scale network simulations, by significantly reducing computational complexity [169]. Meanwhile, edge computing enables decentralized processing of IoT sensor data through lightweight GNN models, reducing latency and cloud dependency. For instance, integrating edge-based GNNs with 5G networks could support real-time leak detection and control actions with sub-50 ms latency [170]. These advancements address scalability and real-time processing challenges, though further research is needed to overcome barriers like quantum error correction and standardized edge AI frameworks for water systems.

From a modeling point of view, GSP may be insufficient to fully capture the intricate information embedded in systems characterized by complex interactions, such as those involved in WDNs. Topological Signal Processing (TSP), which extends GSP by generalizing signal definitions to higher-order spaces, presents a promising direction for future research [171]. The additional structural information encoded in the simplicial complexes may help better understand the multiscale interactions in WDNs. Open issues are related to how TSP can enhance the modeling of flow dynamics, anomaly detection, and network resilience by capturing higher-order dependencies often overlooked by traditional graph-based methods. Furthermore, Topological Deep Learning (TDL) is an emerging area that merges topological insights

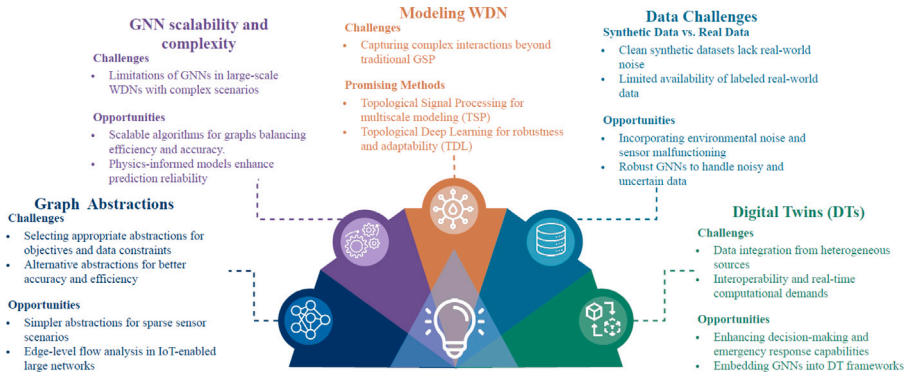


Fig. 8. Future research and open issues summary.

with DL techniques, obtaining global descriptions of multidimensional data and providing robustness to deformation, noise, and adaptability to different types of data [172].

Due to publicly available real-world data scarcity, most methods use synthetic datasets generated via EPANET. These benchmark datasets provide hypothetical infrastructure details like pipe dimensions and reservoir locations. On the one hand, having an ideal dataset represents an advantage for toy model implementation. On the other hand, the analytical methods used to simulate the WDN involve stringent assumptions that inevitably restrict the applicability of analytical results to understanding real-life systems [87]. For example, current numerical models for WDNs are based on the steady-state flow assumption, which assumes a continuous water supply. Indeed, this simplifies the hydraulic equations by neglecting changes over time, providing a snapshot of the system's behavior under fixed conditions. Therefore, they cannot offer realistic evaluations of the state of the elements that make up the WDN during unsteady conditions that are highly dynamic during intermittent water supply [173]. Moreover, it is vital to remember that water operators often do not know the physical structure of WDNs, and data are not labeled properly [67]. Due to the lack of precise information about the actual conditions of the pipes and their precise physical characteristics (which can also evolve), we face significant challenges in synthetic development models that can accurately reflect real-world WDNs. Dealing with uncertainty in the graph structure that models the WDN using Gaussian noise on the edges, such as the one presented in the work by Sabbaqi et al. [174], can lead to superior performance. We, therefore, identify a gap with the real-world scenario that remains a fundamental limitation of state-of-the-art methods.

The reliance on synthetic data introduces another crucial future research direction. Simulated data are typically clean and free from noise or perturbations caused by environmental factors, sensor malfunctions, or measurement inaccuracies. These issues are common in real-world datasets but absent in simulation-generated data. As a result, Current methods, relying on clean synthetic data, fail to address the impact of noise and uncertainty, as real datasets are rarely available. This highlights the need for further research to develop approaches capable of handling data uncertainty [175]. From a city-scaling perspective, it is also unrealistic to assume the same sensor infrastructure across different contexts, as these differences can be caused by other water operators managing the WDN. Even if similar sensors were deployed, calibration issues might arise and erroneous sensor values [176], leading to discrepancies in data accuracy. Moreover, sensors might not be limited to flow and pressure measurements but could include other types, such as acoustic sensors [177]. Therefore, future research should focus on integrating data from different domains and across varying temporal scales to support strategic decisions. One potential solution is the development of pilot deployments like those at the Kuopio Water Cluster.[1] While these may not fully address scalability issues, they could provide smaller, more reliable datasets based on real-world conditions [178,179].

One of the main limitations of current studies on WDSs is their reliance on the steady-state assumption. While this simplification enables tractable modeling and reduces computational complexity, it overlooks the transient phenomena that often characterize real-world hydraulic networks. Events such as valve operations, pump switching, or sudden demand fluctuations can induce short-term dynamic responses, pressure waves, water quality shifts, or flow reversals that are not captured under steady-state conditions. As a result, models trained on steady-state data may fail to detect fast-developing anomalies or misrepresent the system's true operational state. A promising direction for future research lies in integrating transient-based methods [180]. Unlike steady-state approaches, transient analysis captures the dynamic behavior of hydraulic variables over time, offering a more detailed and responsive system representation. This temporal resolution can provide richer signals for tasks such as leak detection and state estimation, especially when dealing with fast-changing events or anomalies that may be missed under steady-state assumptions. Recent research is making transient-based methods increasingly viable. Song et al. [181] proposes to extend the Smoothed Particle Hydrodynamics (SPH) method to solve the transient equations of more complex WDSs. Based on the discretization of the fluid as a set of particles, SPH could help capture complex flow phenomena such as column separation, slug flows, and rapid filling in practical scenarios. Tjuatja et al. [182] proposes a Physics Informed Neural Network that demonstrated high accuracy in reconstructing transient pressure signals under uncertainty, outperforming classical models, though at the cost of higher computational effort. A similar approach has been given by Yan et al. [183], that introduce a hybrid model that integrates hydraulic transient laws with physics-informed neural networks embedding nodal hydraulic behavior into the loss function, enabling the entire network to be trained as a single neural structure. The model is robust in handling sparse and noisy sensor data while accurately predicting pressure and flow transients across various operational scenarios, including burst events and pump shutdowns.

Another emerging trend is the creation of virtual representations of WDNs, commonly called Digital Twins (DTs), which offer new opportunities for real-time monitoring and system optimization. Among the existing frameworks, the Asset Administration Shell (AAS), a core component of the Reference Architecture Model for Industry 4.0 (RAMI 4.0), offers a standardized approach to structuring and managing the digital representation of physical assets [184]. The potential benefits include improved decision-making through simulation of operational scenarios, energy optimization, and enhanced emergency response capabilities [185]. In this scenario, it is possible to work with real data integrated into specific points of sensor information. However, several challenges must be addressed, such as data integration from heterogeneous sources, interoperability with existing systems, and the computational demands of real-time simulations [186]. It is essential to understand how GNNs fit existing DT frameworks and how they can be integrated with the core components that are being standardized.

[1] https://kuopiowatercluster.com/services/.

8. Conclusions

Global water consumption is on a steady rise due to population growth and urbanization, experiencing a substantial annual increase in demand, juxtaposed with a looming global water shortage predicted to reach about 40% by 2030 [3]. Information and Communication Technologies (ICTs), particularly the Internet of Things (IoT), combined with advanced, real-time data analysis, can lead to the digitization of WDNs and help increase water usage efficiency. In particular, advanced AI techniques like GNNs have demonstrated substantial potential in analyzing the data collected from sensing devices to address key challenges in WDNs, such as leak detection, state estimation, demand forecasting, optimal sensor placement, and water quality monitoring.

In this paper, we have reviewed some very recent and characteristic studies that apply GNN techniques to address a specific challenge in WDNs. The results surveyed demonstrate that GNNs have the potential to effectively leverage network topology to model and optimize the operation of WDNs, providing significant advantages over traditional methods. The integration of physical laws, even when sparse or incomplete measurements are available, allows the design of models that achieve adequate levels of accuracy and show robust performance under various operating conditions.

Specifically, the studies reviewed highlight several critical challenges and opportunities in applying GNNs to WDN management. Key challenges include scalability and complexity in large-scale WDNs, the need for robust graph abstractions, and the integration of heterogeneous data sources into digital twin frameworks. However, significant opportunities exist, such as developing scalable algorithms that balance efficiency and accuracy, incorporating physics-informed models to enhance prediction reliability, and using simpler or alternative graph abstractions for sparse sensor scenarios. Additionally, advancements in TSP and TDL offer promising methods to capture complex interactions within WDNs. Addressing these challenges and leveraging these opportunities can lead to more robust, adaptive, and efficient solutions for real-time WDN management, ultimately supporting sustainable water usage amid growing global demand.

The research direction presented here is rich in future research directions. Investigating alternative graph representations of the physical WDNs to design the GNN has significant potential for developing more robust models and achieving higher accuracy. Considering the different sensing elements that can be deployed as part of the IoT infrastructure and, consequently, the data richness that can be acquired while monitoring the operation of the WDN, there is a need to extend the existing GNN architectures to encode the resulting multimodal representation of the WDN elements. The evolution of fundamental GNN techniques will offer new perspectives and may help expand the use of GNNs for modeling and optimizing the operation of WDNs.

CRediT authorship contribution statement

Giacomo Vittori: Writing – original draft, Methodology, Investigation, Formal analysis. **Yelizaveta Falkouskaya:** Writing – original draft, Investigation, Formal analysis. **Daniel M. Jimenez-Gutierrez:** Writing – review & editing, Investigation, Conceptualization. **Tiziana Cattai:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis. **Ioannis Chatzigiannakis:** Writing – review & editing, Writing – original draft, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

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Data availability

No data was used for the research described in the article.

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